



COURSE SLIDES · WSQ

Fundamentals of AI Ethics and Responsible AI

Course Code: TGS-2023021102

TSC: AI Ethics and Governance · ICT-CGP-2026-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 7.0 · 9 September 2026

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COURSE ADMINISTRATION

00

Welcome and Course Orientation

Expectations · learning outcomes · assessment · course access

Digital Attendance (Mandatory)

1 AM attendance

Scan the SSG digital-attendance QR code when the trainer displays it.

TRAQOM

2 PM attendance

Repeat after lunch; confirm the submission screen before closing.

TRAQOM

3 Assessment attendance

Complete the separate assessment attendance at the scheduled time.

TRAQOM

4 Funding requirement

Maintain at least 75% attendance and complete all prescribed assessments.

TRAQOM

THE POINT

TRAQOM attendance is a formal funding record—check each submission before continuing.

About the Trainer



Your Trainer

Complete before class

NAME

TITLE / ROLE

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Technology educator · AI governance
and responsible innovation

NAME

Dr Alfred Ang

TITLE / ROLE

Principal Trainer

QUALIFICATIONS

PhD · adult education · technology

AREAS OF EXPERTISE

AI, governance, data and digital transformation

TRAINING EXPERIENCE

WSQ and professional technology programmes

CONTACT

enquiry@tertiaryinfotech.com

Let's Know Each Other

1 Name

How would you like the class to address you?

INTRODUCTIONS

2 Role

Where do AI-assisted decisions appear in your work?

INTRODUCTIONS

3 Experience

Which AI tools or models have you used?

INTRODUCTIONS

4 Goal

What governance decision do you want to make better?

INTRODUCTIONS

THE POINT

Listen for different perspectives: AI affects teams and stakeholders differently.

Ground Rules

1 Participate

Questions and respectful challenge improve the class.

HOUSEKEEPING

2 Protect data

Use only supplied mock data; never paste confidential information.

HOUSEKEEPING

3 Be evidence-led

Separate facts, assumptions, values and recommendations.

HOUSEKEEPING

4 Stay human-centric

Discuss impacts on people, not only model performance.

HOUSEKEEPING

THE POINT

A psychologically safe class supports rigorous ethical reasoning.

Course at a Glance

1 Programme

WSQ · beginner level ·
instructor-led and hands-on.

COURSE FACTS

2 Duration

2 days · 16 total contact hours
· includes 2 hours of
assessment.

COURSE FACTS

3 TSC

AI Ethics and Governance ·
ICT-CGP-2026-1.1.

COURSE FACTS

4 Assessment

Written Assessment (SAQ) +
Case Study · open book.

COURSE FACTS

THE POINT

The course moves from principles to a defensible governance decision.

Course Proposal Alignment

1 LU1 · LO1

Identify AI ethics and governance principles · K1 frameworks · K2 principles · A1 identify principles and processes.

CONSTRUCTIVE

2 LU2 · LO2

Apply an AI ethics and governance framework to AI projects · K3 processes · K4 code · A2 uphold · A3 govern deployment.

CONSTRUCTIVE

3 Training allocation

LU1: Day 1, 09:00–17:00 excluding lunch = 7 hours ·
LU2: Day 1, 17:00–18:00 plus
Day 2, 09:00–16:00 excluding lunch = 7 hours.

CONSTRUCTIVE

4 Assessment

WA Q1–Q4 assess K1–K4 ·
Case Study Q1–Q3 assess A1–A3 · 2 hours total.

CONSTRUCTIVE

THE POINT

The approved Course Proposal maps every knowledge and ability statement to teaching, practice and assessment.

TSC Knowledge and Abilities

1 K1–K2

Frameworks and principles for AI ethics and governance.

TSC MAPPING

2 K3–K4

Governance processes and relevant codes of conduct.

TSC MAPPING

3 A1

Identify principles and processes.

TSC MAPPING

4 A2–A3

Uphold governance duties and apply them to AI projects.

TSC MAPPING

THE POINT

The activity sequence provides direct practice for every K and A code.

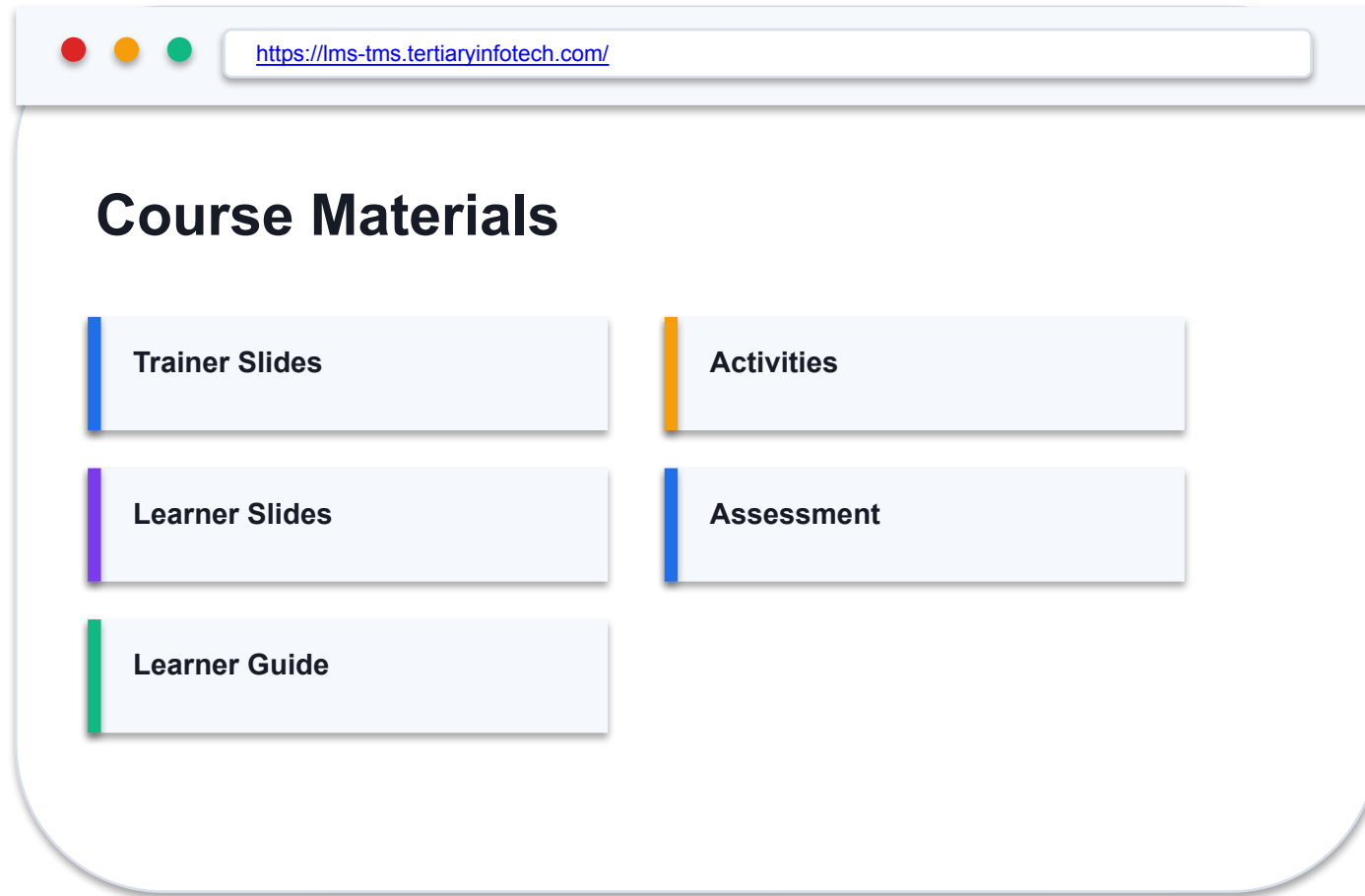
Lesson Plan · Day 1

09:00–09:30	Administration, introductions and AM attendance
09:30–10:30	Module 1 · AI Ethics Foundations
10:30–10:40	Tea break
10:40–11:25	Activity 1 · Use-case and stakeholder map
11:25–12:25	Module 2 · Responsible AI Principles
12:25–13:10	Activity 2 · AI Code of Conduct
13:10–14:10	Lunch break
14:10–14:50	Module 3 · AI Governance Frameworks
14:50–15:50	Activity 3 · Risk tiering and RACI
15:50–16:00	Tea break
16:00–17:00	Module 4 · AI Governance Lifecycle
17:00–17:50	Activity 4 · Impact assessment and risk register
17:50–18:00	Day 1 recap and PM attendance

Lesson Plan · Day 2

09:00–09:10	Day 1 recap and AM attendance
09:10–09:50	Module 5 · Fairness and Explainability
09:50–10:00	Tea break
10:00–11:00	Activity 5 · Fairness and explainability audit
11:00–11:35	Module 6 · Data Governance and Privacy
11:35–12:20	Activity 6 · Data governance controls
12:20–13:20	Lunch break
13:20–13:50	Module 7 · Generative AI Security and Copyright
13:50–14:00	Tea break
14:00–14:50	Activity 7 · GenAI threat model
14:50–15:20	Module 8 · Human Oversight and Accountability
15:20–15:50	Activity 8 · Oversight and governance decision
15:50–16:00	Assessment briefing and PM attendance
16:00–17:00	Written Assessment (SAQ)
17:00–18:00	Case Study assessment

Course Material Hub



1

Current approved courseware

2

Learner Guide and slide deck

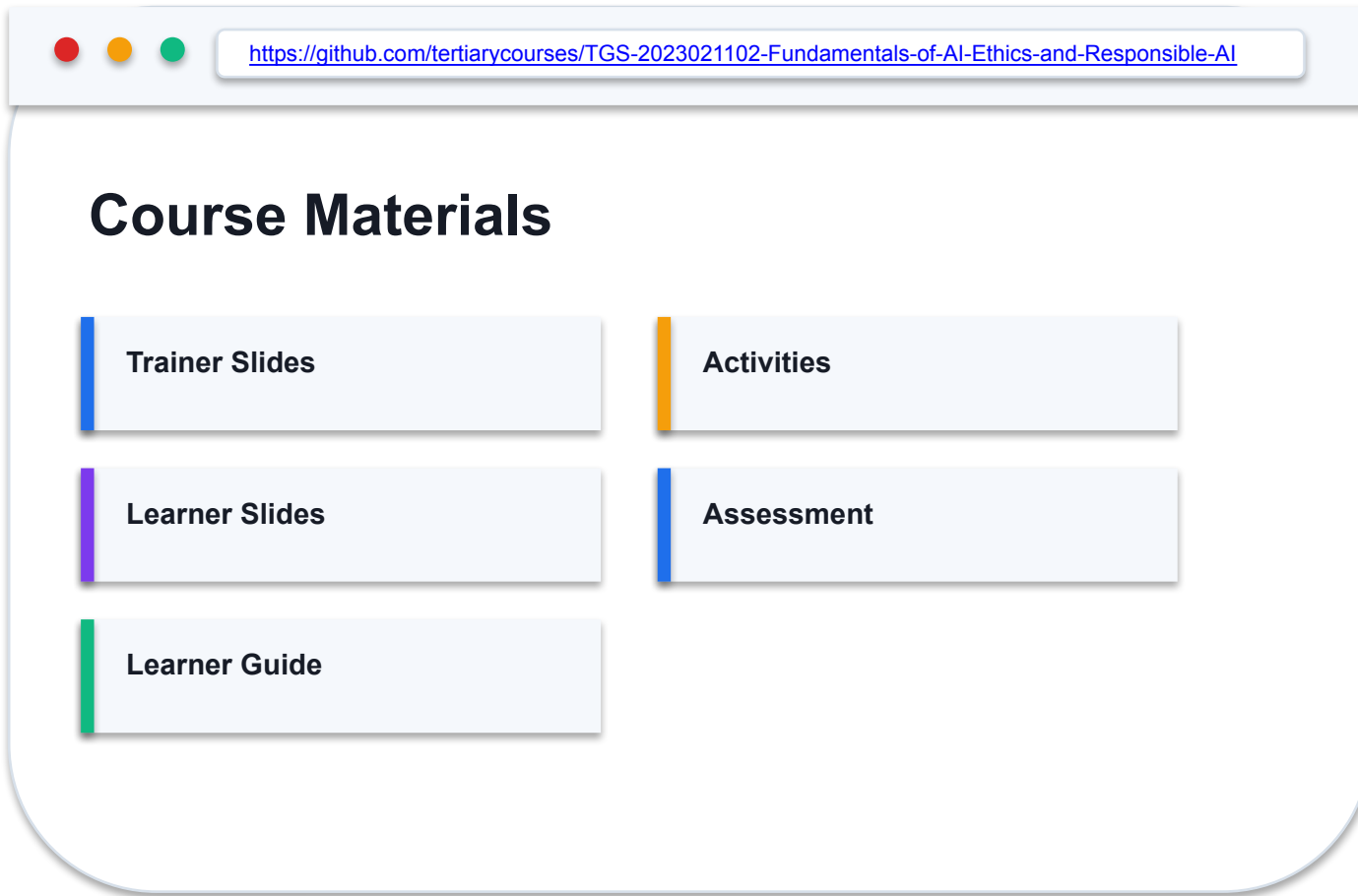
3

Assessment submission area

4

Attendance and course records

Hands-On Activity Library



1

Eight self-contained activity folders

2

Prompt pack for each activity

3

Supplied mock data and templates

4

Submission and acceptance checklist

Briefing for Assessment

1 Materials

Open book: approved slides and Learner Guide only.

ASSESSMENT BR

2 Conduct

No discussion, photos, recording or unauthorised assistance.

ASSESSMENT BR

3 Submission

Complete answers in the provided documents and upload to LMS.

ASSESSMENT BR

4 Competency

All required K and A criteria must be demonstrated.

ASSESSMENT BR

THE POINT

Ask for clarification before the assessment begins; follow the assessor's instructions.

Final Assessment

1 WA (SAQ)

4 open-ended questions · K1–K4 · 1 hour.

ASSESSMENT

2 Case Study

3 open-ended questions · A1–A3 · 1 hour.

ASSESSMENT

3 Format

Open book using approved course materials.

ASSESSMENT

4 Appeal

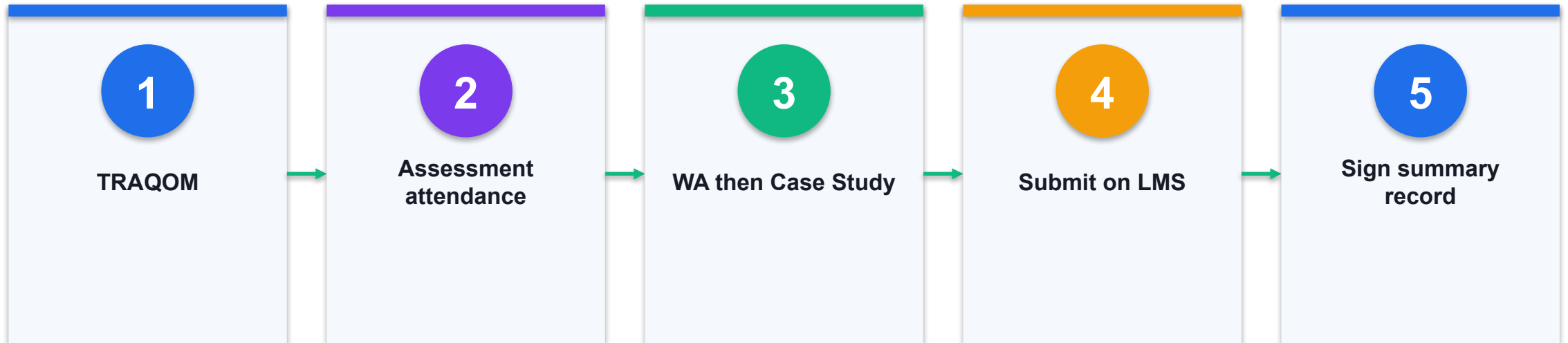
The assessor explains feedback, sign-off and appeal rights.

ASSESSMENT

THE POINT

The assessment mirrors the approved instrument: 4 WA questions and 3 Case Study questions.

Assessment Flow



Complete the steps in sequence and confirm each submission.

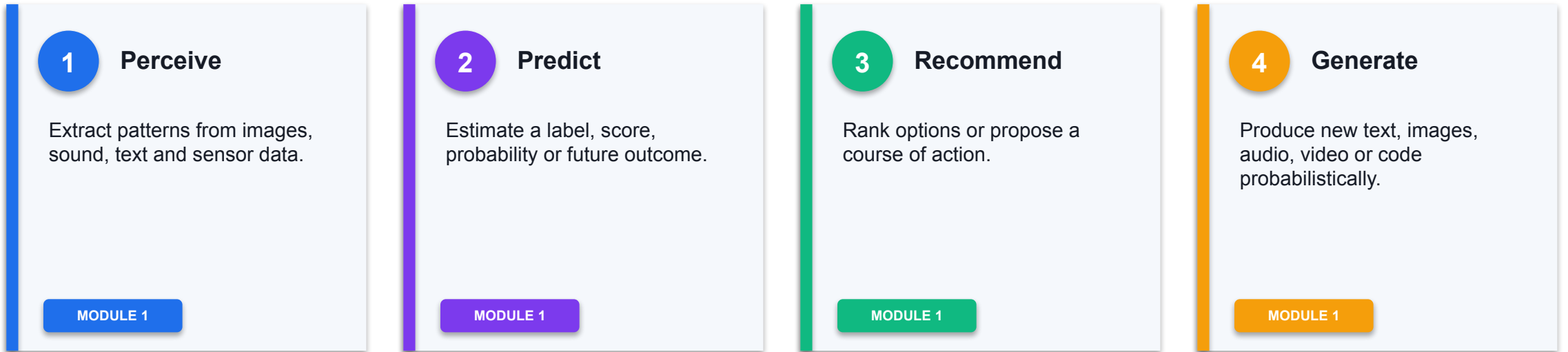
MODULE 1 · K1

01

AI Ethics Foundations

From AI capability to sociotechnical impact

What AI systems actually do



THE POINT

Ethical risk follows the decision and impact, not the marketing label attached to the model.

Why AI ethics is operational

1 Scale

A single defect can affect thousands of people before anyone notices.

MODULE 1

2 Opacity

Complex pipelines can hide why a result changed or who is accountable.

MODULE 1

3 Power

AI can shift opportunities, burdens and decision authority.

MODULE 1

4 Feedback

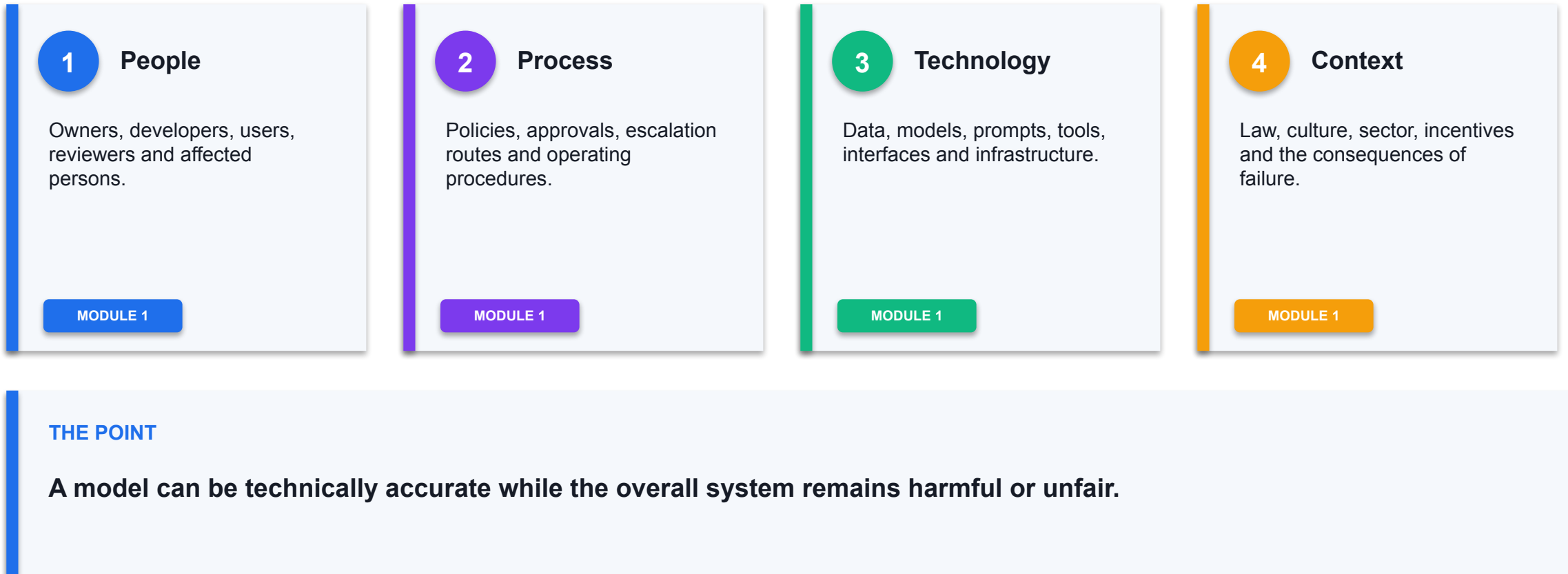
Predictions influence behaviour and create the next training data.

MODULE 1

THE POINT

AI ethics is how an organisation turns values into decisions, controls, evidence and redress.

The AI system is sociotechnical



Risk, harm and benefit

1 Risk

Likelihood multiplied by severity, adjusted for exposure and control strength.

MODULE 1

2 Harm

Impact on rights, safety, opportunity, dignity, privacy or the environment.

MODULE 1

3 Benefit

Value to individuals, organisations and society—not only efficiency.

MODULE 1

4 Trade-off

A justified balance with documented residual risk and accountable approval.

MODULE 1

THE POINT

Responsible AI seeks proportionate controls, not zero risk and not innovation at any cost.

Three ethical lenses

1 Consequences

Which option produces the greatest overall benefit and least harm?

MODULE 1

2 Duties and rights

What obligations and non-negotiable rights must be respected?

MODULE 1

3 Virtues and values

What would a trustworthy, fair and responsible organisation do?

MODULE 1

4 Public reason

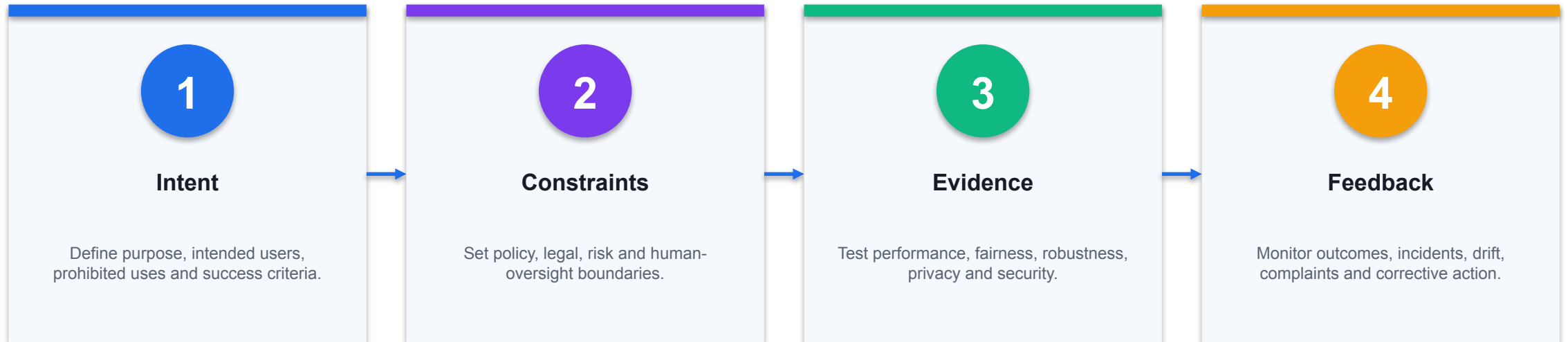
Could the decision and rationale be explained to affected people?

MODULE 1

THE POINT

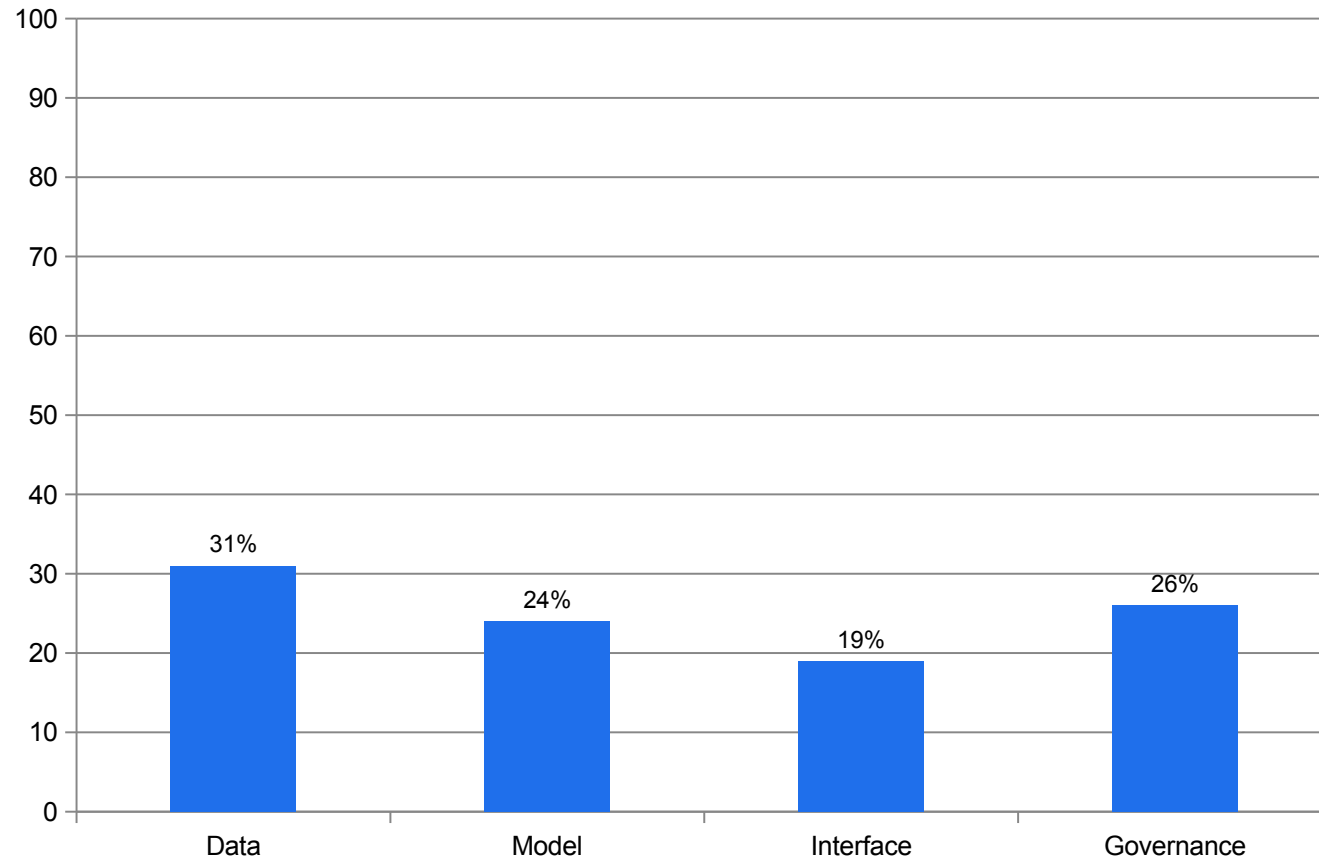
Using more than one lens exposes blind spots that a single moral theory can miss.

Responsible AI as a control system



A principle becomes real only when it has an owner, control, metric and evidence trail.

AI incidents have multiple causes



Data

Unrepresentative samples, weak labels, leakage or unlawful collection.

Model

Poor calibration, brittleness, opacity or unsafe generalisation.

Interface

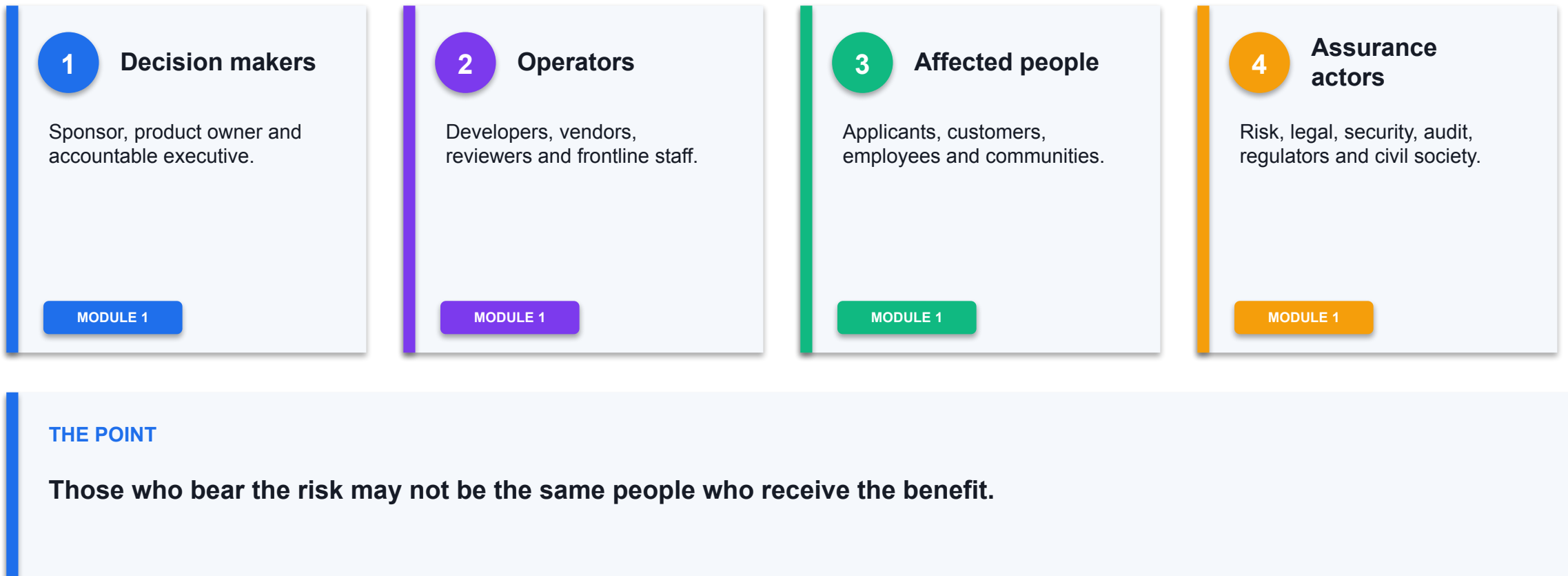
Automation bias, misleading confidence or no meaningful choice.

Governance

No owner, weak monitoring, unclear escalation or ignored warnings.

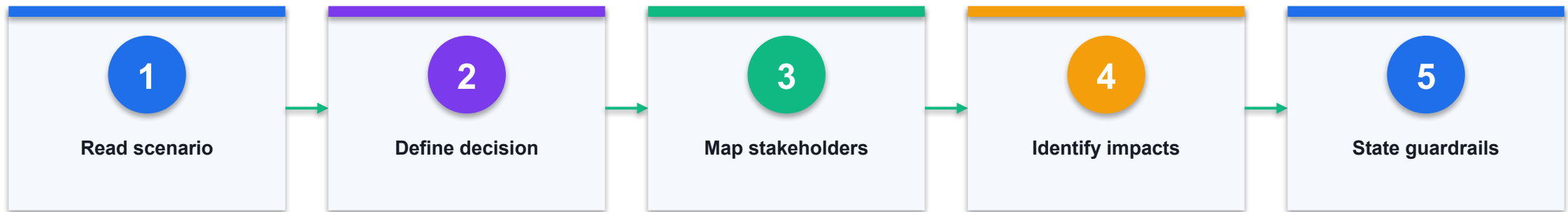
Illustrative teaching data shows that non-model causes can account for most incident pathways.

Stakeholder impact map



Map an AI Use Case and Its Stakeholders

Identify the system purpose, decision, stakeholders, benefits, harms and ethical tensions (A1).



YOU'LL PRODUCE Completed AI Use-Case & Stakeholder Canvas

Tools: Prompt PDF, stakeholder template, scenario brief · Detailed procedure: Learner Guide, Activity 1

Verify · Map an AI Use Case and Its Stakeholders

PASS WHEN

The canvas names the decision, at least six stakeholders, two benefits, four harms, three prohibited uses and an accountable owner.



AI-assisted decision is explicit



Affected people are separate from users



Benefits and harms name who experiences them



Assumptions are labelled



Prohibited uses and owner are present

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

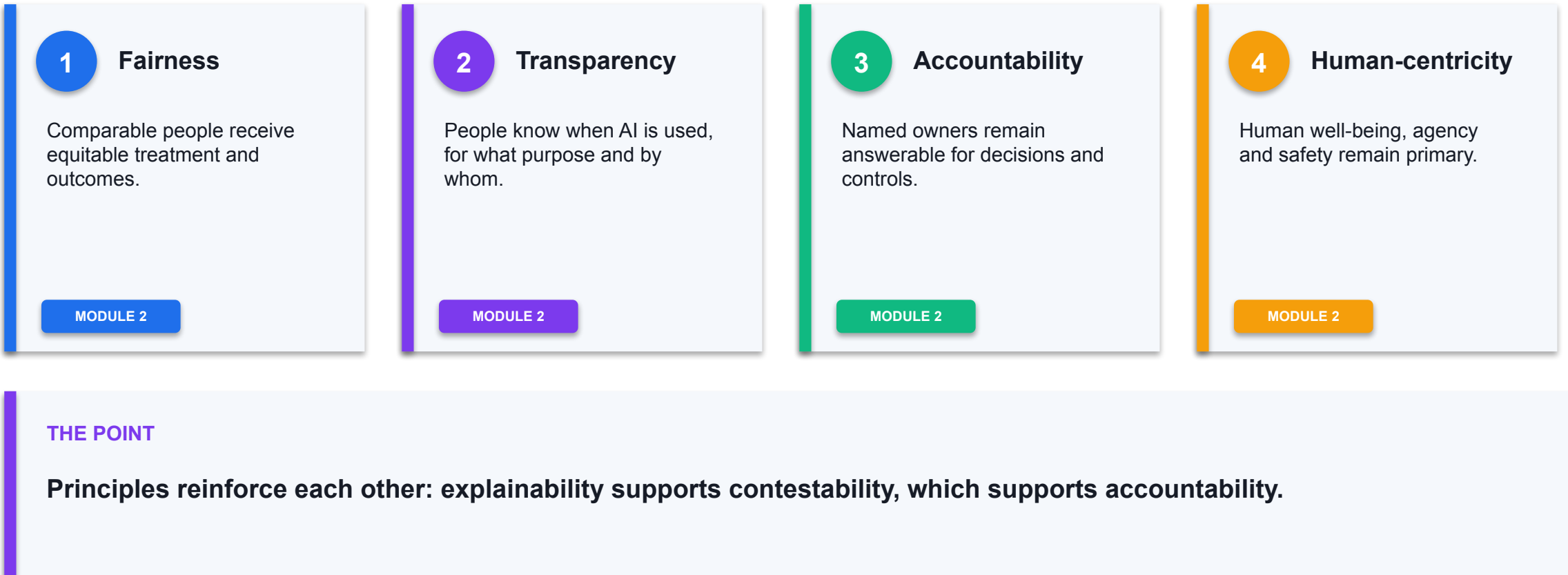
MODULE 2 · K2

02

Responsible AI Principles

Translate values into observable behaviour

Principles form a connected system



Fairness is a design decision

1 Define groups

Select protected or vulnerable groups relevant to the context.

MODULE 2

2 Define outcome

State what favourable and harmful outcomes mean.

MODULE 2

3 Choose metric

Pick a fairness measure that matches the decision pathway.

MODULE 2

4 Accept trade-offs

Document who benefits, who is burdened and why.

MODULE 2

THE POINT

There is no universal fairness metric; the defensible choice begins with context and harm.

Transparency has layers

1 System notice

Disclose that AI is involved and what role it plays.

MODULE 2

2 Process clarity

Explain inputs, workflow, controls and decision authority.

MODULE 2

3 Technical evidence

Provide model cards, data sheets, tests and audit logs.

MODULE 2

4 Outcome reason

Give the affected person a useful reason and next step.

MODULE 2

THE POINT

Transparency is audience-specific: a user needs clarity; an auditor needs traceable evidence.

Explainability, interpretability, transparency

1 Interpretability

How directly a human can understand the model mechanism.

MODULE 2

2 Explainability

A post-hoc account of factors influencing a result.

MODULE 2

3 Transparency

Information about purpose, data, process, limits and ownership.

MODULE 2

4 Contestability

A practical way to question, correct or appeal a decision.

MODULE 2

THE POINT

An explanation is only useful when it supports an action, not merely curiosity.

Privacy and data stewardship

1 Purpose

Collect and use data for clear, legitimate and compatible purposes.

MODULE 2

2 Minimisation

Use only what is necessary for the stated objective.

MODULE 2

3 Protection

Apply access, encryption, retention and disclosure controls.

MODULE 2

4 Agency

Enable notice, consent where appropriate, correction and withdrawal.

MODULE 2

THE POINT

Privacy is a lifecycle obligation, not a one-time consent checkbox.

Safety, security and robustness

1 Safety

Prevent unacceptable physical, psychological or societal harm.

MODULE 2

2 Security

Protect confidentiality, integrity and availability against threats.

MODULE 2

3 Robustness

Maintain acceptable behaviour under variation and attack.

MODULE 2

4 Resilience

Detect failure, contain impact and recover safely.

MODULE 2

THE POINT

Security protects the system; safety protects people; responsible AI requires both.

Accountability needs evidence

1 Owner

A named role accepts responsibility for the system and outcome.

MODULE 2

2 Decision record

Approvals, assumptions, exceptions and residual risk are logged.

MODULE 2

3 Assurance

Independent review tests whether controls work as claimed.

MODULE 2

4 Remedy

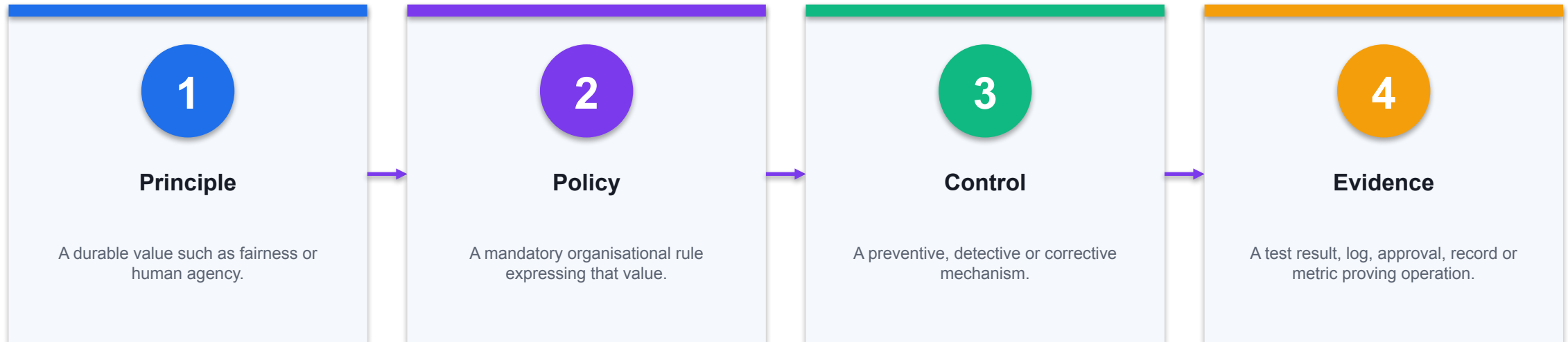
Affected people have a route to correction, appeal and redress.

MODULE 2

THE POINT

If nobody can show the decision record, accountability exists only on paper.

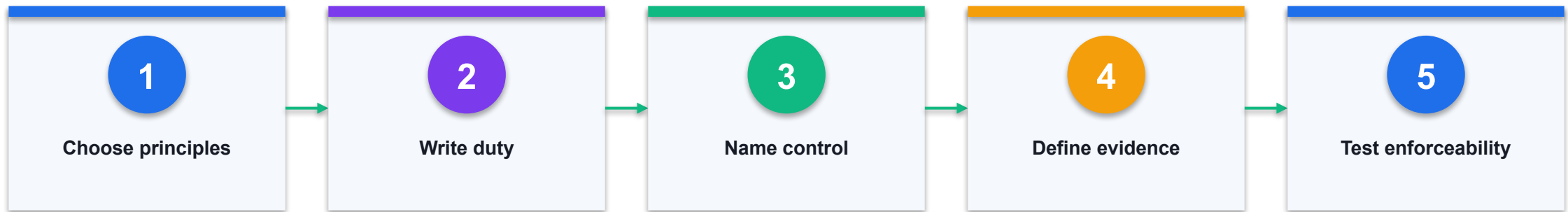
Principles to policy to practice



Write every code-of-conduct statement so that compliance can be observed and audited.

Turn Principles into an AI Code of Conduct

Translate responsible AI principles into observable duties, controls and evidence (A1, A2).



YOU'LL PRODUCE AI Ethics Code of Conduct with control statements

Tools: Prompt PDF, policy template, principle-control matrix · Detailed procedure: Learner Guide, Activity 2

Verify · Turn Principles into an AI Code of Conduct

PASS WHEN

Every principle has a mandatory duty, owner, control, frequency/trigger, evidence and failure indicator.



Statements use must, not should



Each control has one owner



Evidence is named



Exception route is documented



Human agency and redress are covered

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

COURSE PAUSE

00

Lunch Break

45 minutes

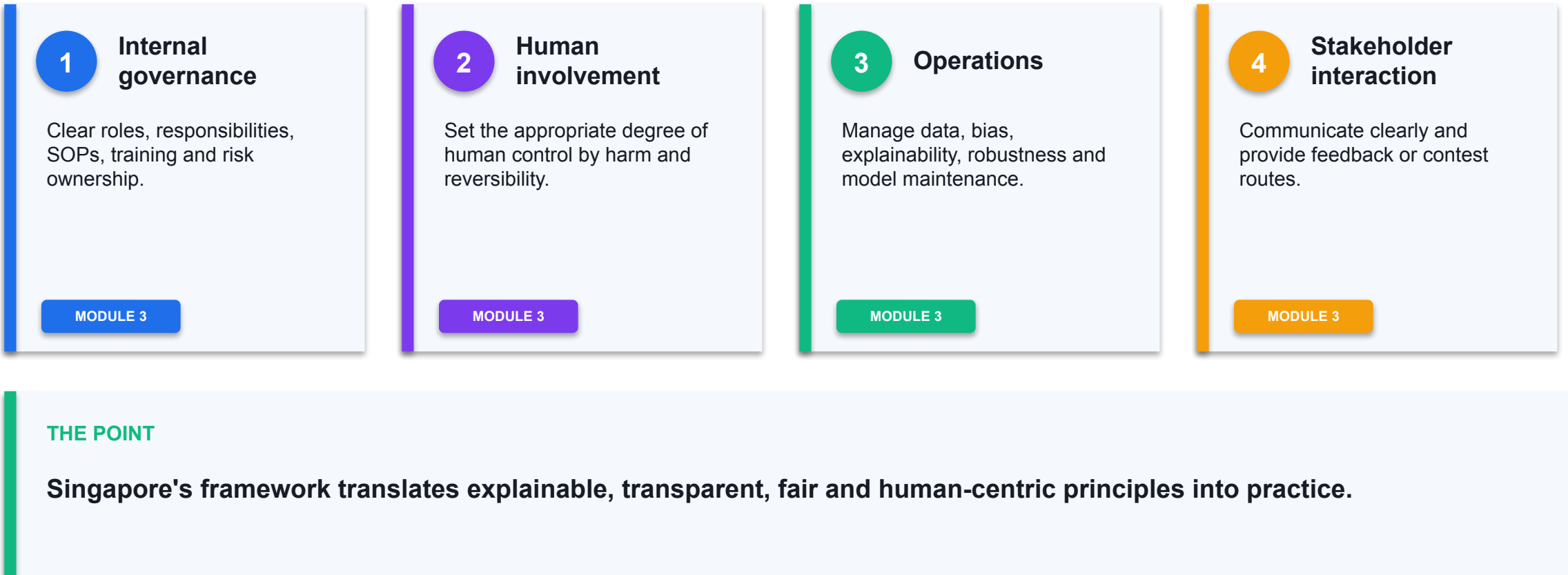
MODULE 3 · K1 · K2 · A1

03

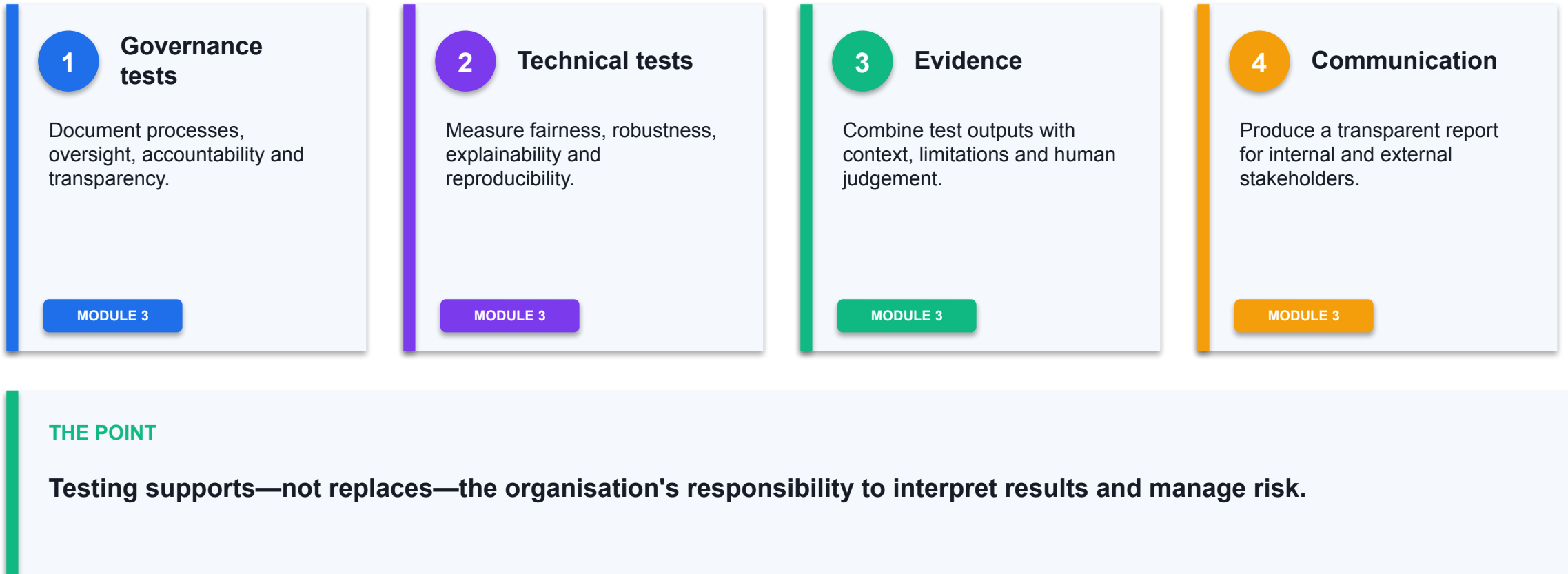
AI Governance Frameworks

Use complementary frameworks without creating checklist theatre

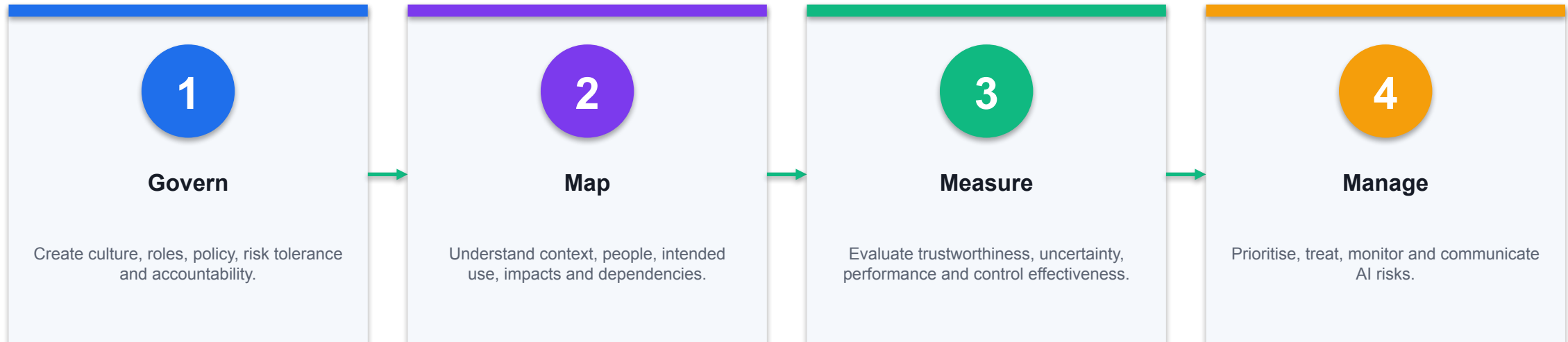
Singapore Model AI Governance Framework



AI Verify assurance model

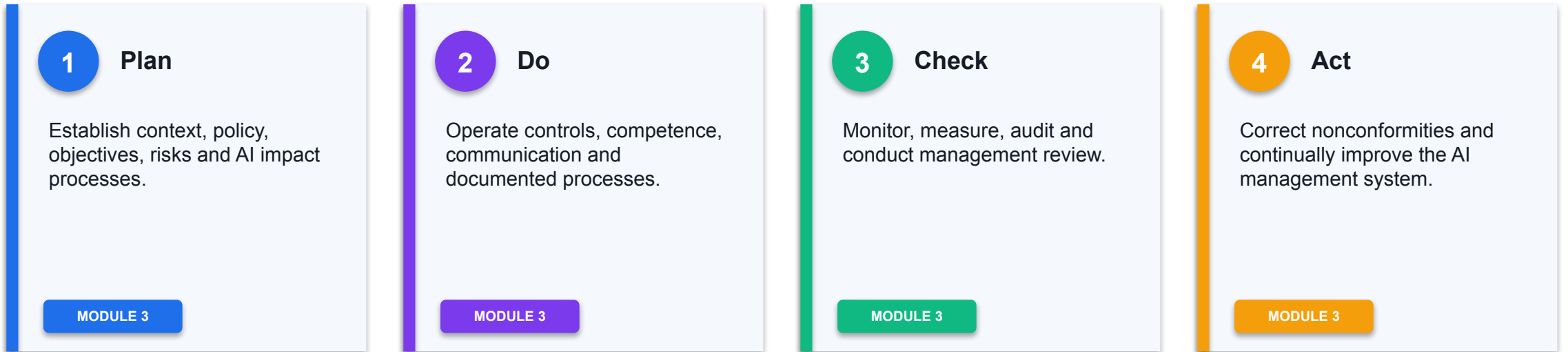


NIST AI RMF core



Govern is cross-cutting; Map, Measure and Manage iterate throughout the lifecycle.

ISO/IEC 42001 management system



THE POINT

ISO 42001 governs the organisational system around AI, not the technical quality of one model alone.

Frameworks serve different questions

1 Singapore MGF

How should an organisation operationalise trustworthy AI?

MODULE 3

2 AI Verify

How can governance and technical claims be tested and evidenced?

MODULE 3

3 NIST AI RMF

How do we structure continuous AI risk management?

MODULE 3

4 ISO 42001

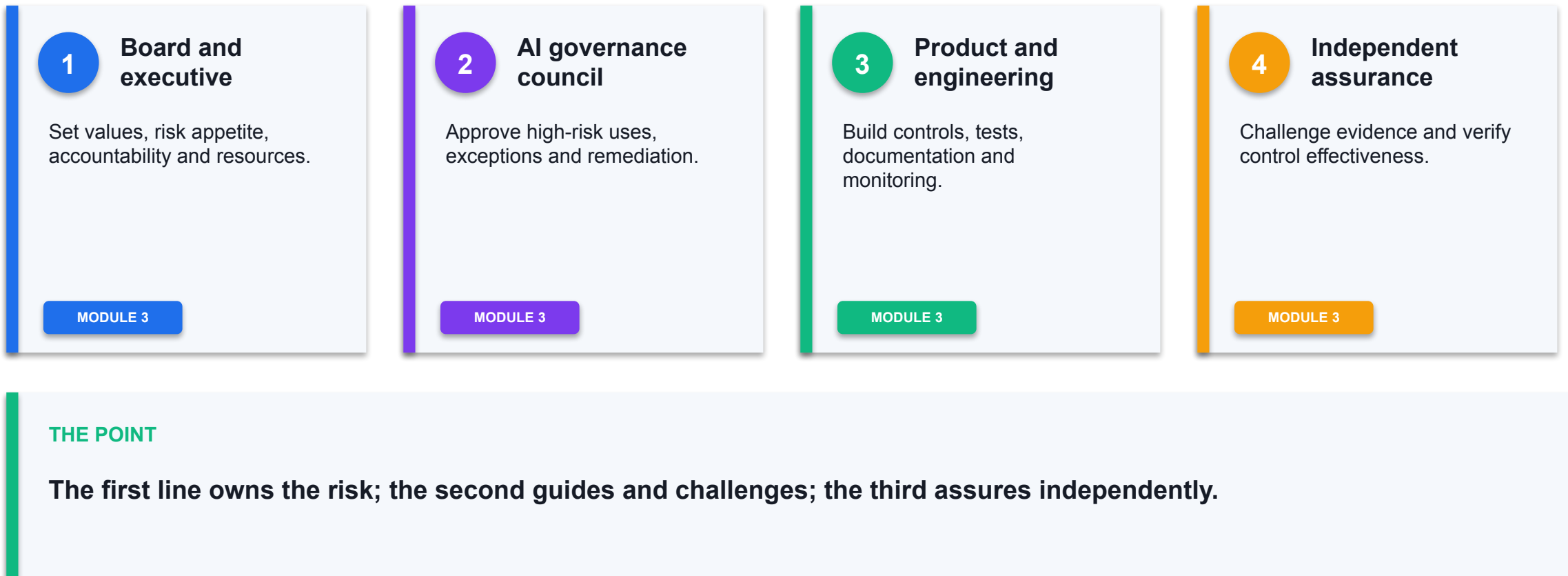
How do we manage and continually improve an enterprise AIMS?

MODULE 3

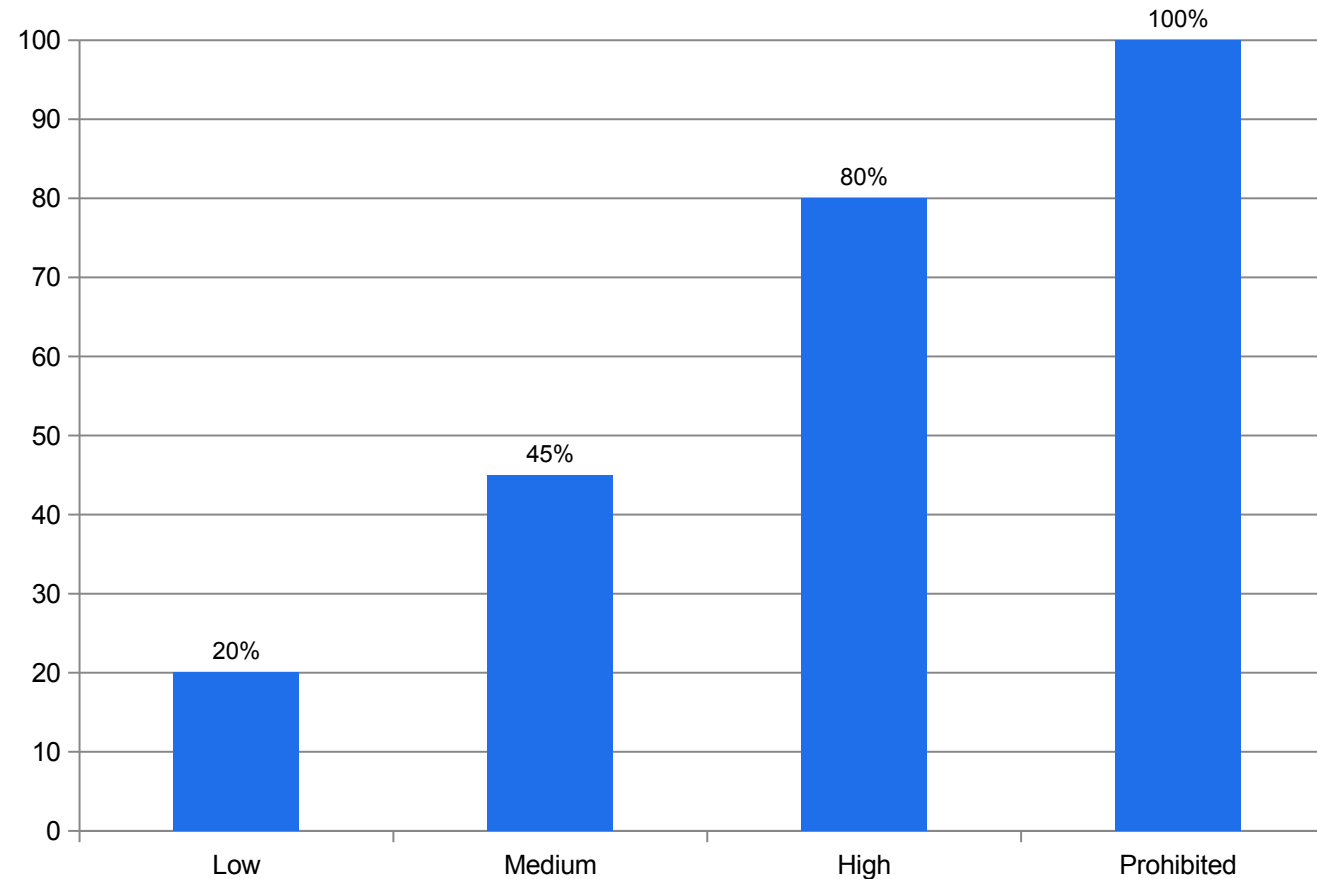
THE POINT

Crosswalk overlapping requirements once; do not make teams complete four disconnected checklists.

AI governance operating model



Use-case inventory and risk tiers



Inventory

Record owner, purpose, users, data, model, vendor and deployment state.

Inherent risk

Assess impact before controls using harm, scale and vulnerability.

Control profile

Match testing, approvals and oversight to the risk tier.

Residual risk

Approve, reduce, transfer, avoid or monitor after controls.

Control intensity rises with inherent risk; prohibited uses are stopped rather than mitigated into approval.

Governance anti-patterns

1 Policy-only

A principle document exists but product teams lack controls and evidence.

MODULE 3

2 Committee bottleneck

Every decision waits for one central group regardless of risk.

MODULE 3

3 Vendor blind trust

Contract language substitutes for technical and operational assurance.

MODULE 3

4 Launch-and-forget

Approval occurs once with no monitoring, incident route or review trigger.

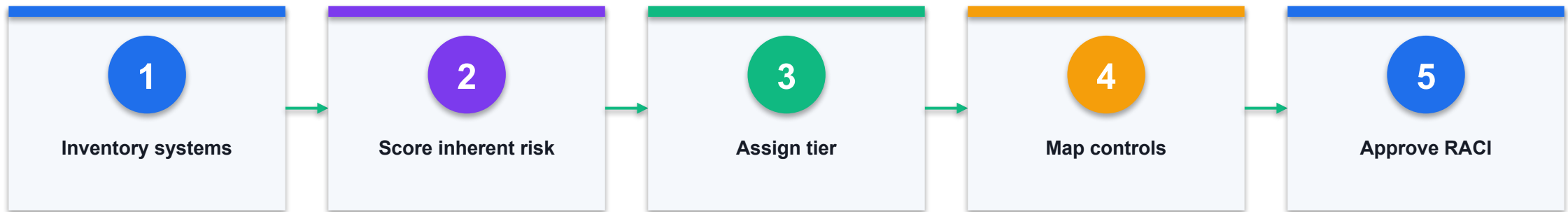
MODULE 3

THE POINT

Effective governance is risk-based, distributed, evidence-led and continuous.

Tier AI Risk and Build a Governance RACI

Apply governance frameworks to classify AI use cases and assign decision rights (A1, A3).



YOU'LL PRODUCE AI inventory, risk-tier rationale and governance RACI

Tools: Prompt PDF, AI inventory CSV, RACI template · Detailed procedure: Learner Guide, Activity 3

Verify · Tier AI Risk and Build a Governance RACI

PASS WHEN

All use cases have a defensible tier, minimum control profile, review cadence and unambiguous RACI.



Inventory is complete



Scores have reasons



Prohibited-use rule exists



One Accountable role per decision



Framework controls are crosswalked

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

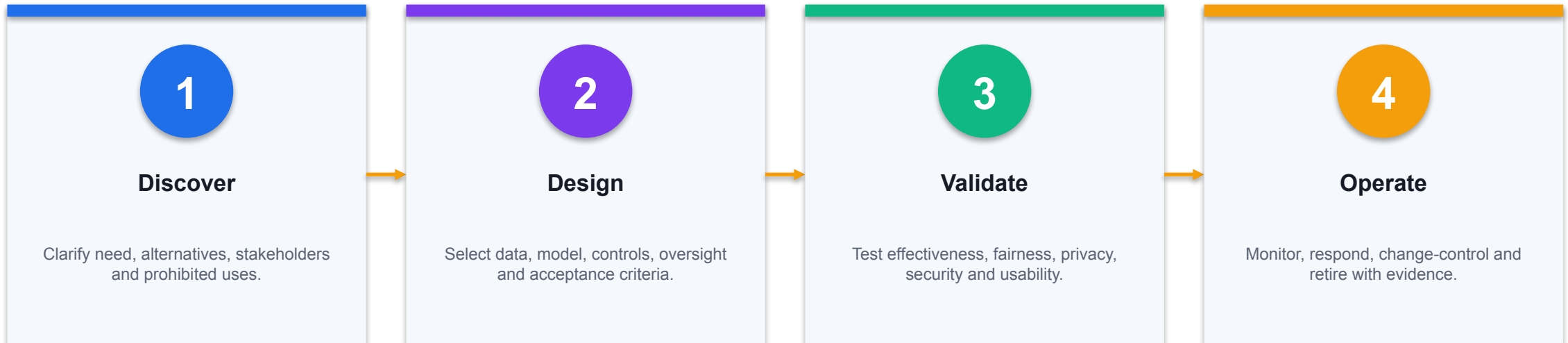
MODULE 4 · K3 · A1 · A3

04

AI Governance Lifecycle

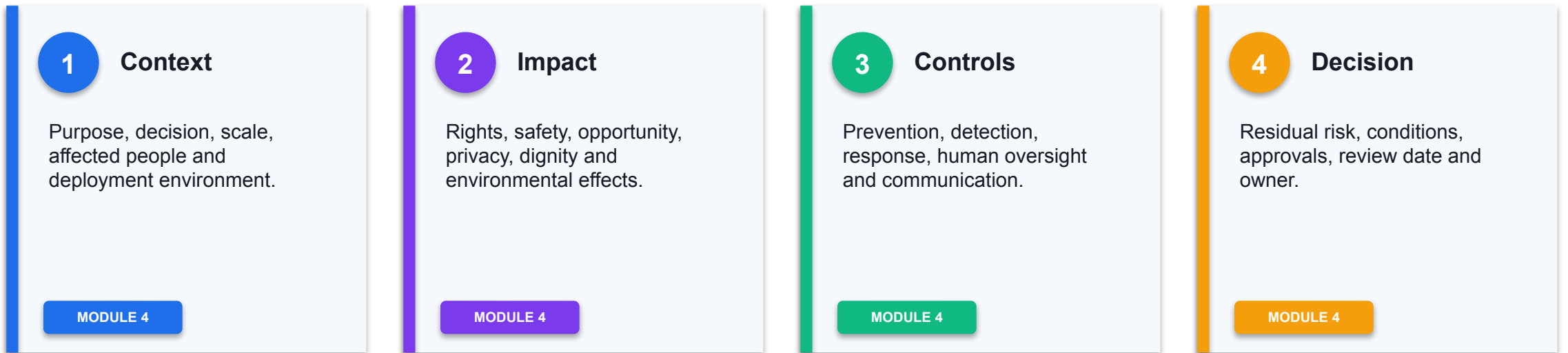
Control the system from idea to retirement

Lifecycle governance gates



A gate is a decision with entry criteria, evidence, accountable approver and recorded outcome.

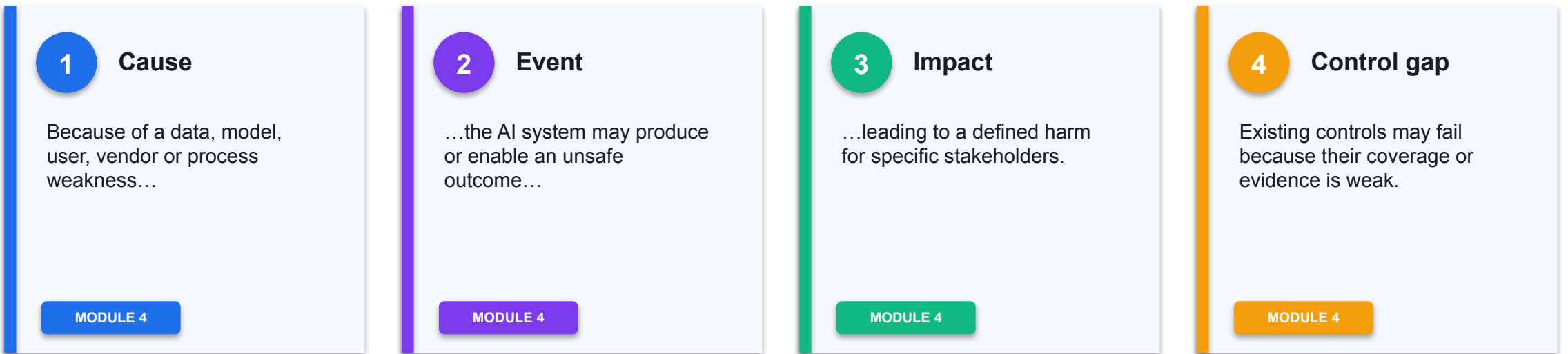
AI impact assessment



THE POINT

The assessment should change the design; a form completed after launch is not risk management.

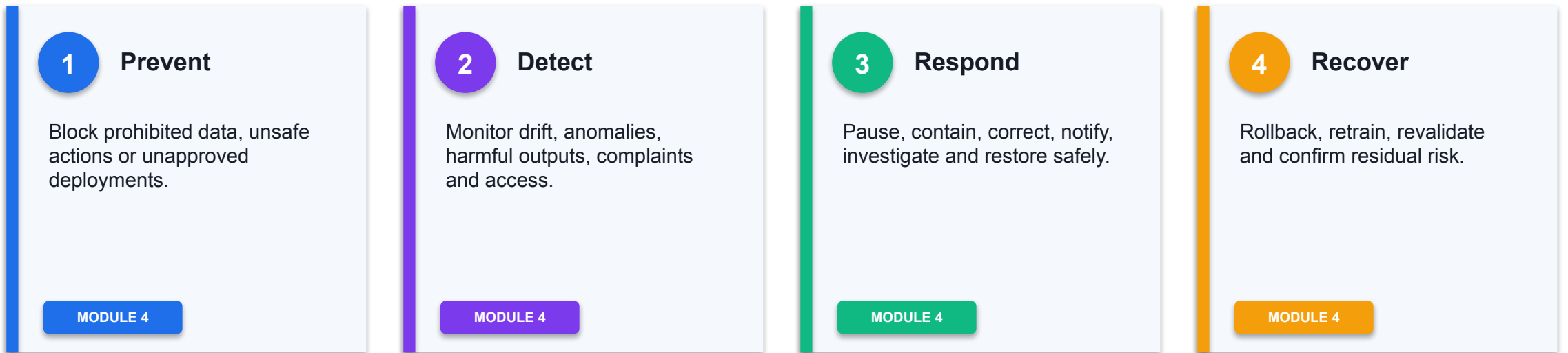
Risk statement anatomy



THE POINT

Specific risk statements lead to testable controls; vague labels such as 'bias risk' do not.

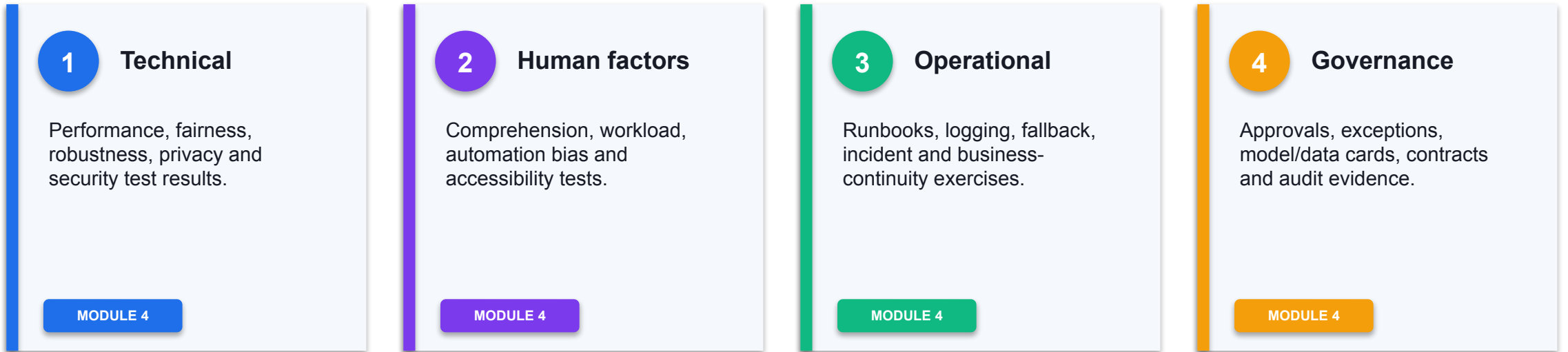
Control design



THE POINT

Use defence in depth; a single filter or reviewer is a fragile point of failure.

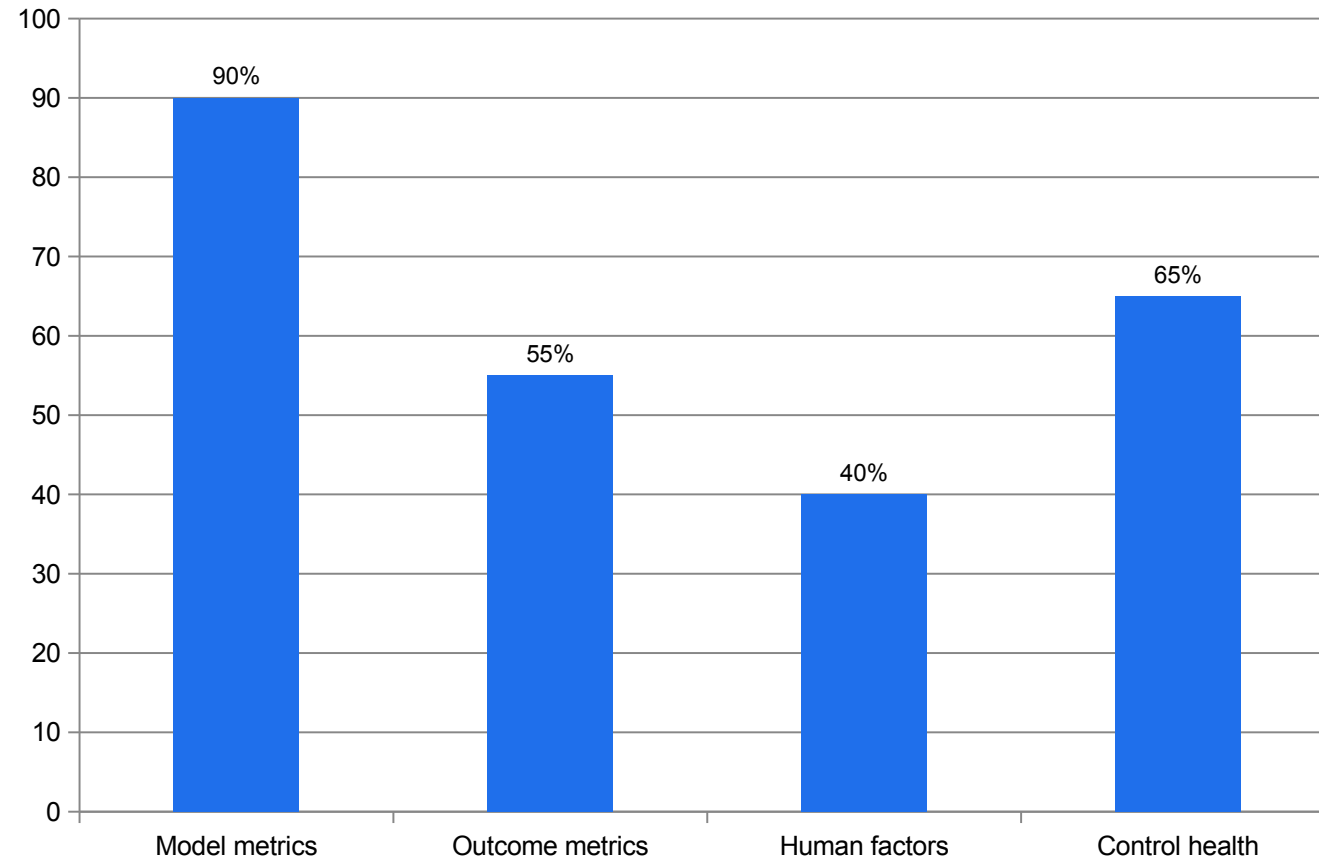
Validation evidence stack



THE POINT

A benchmark score is necessary evidence, but never sufficient evidence for responsible deployment.

Production monitoring



Inputs

Population shift, missingness, sensitive data and prompt patterns.

Outputs

Error, toxicity, fairness gaps, confidence and unsafe actions.

Outcomes

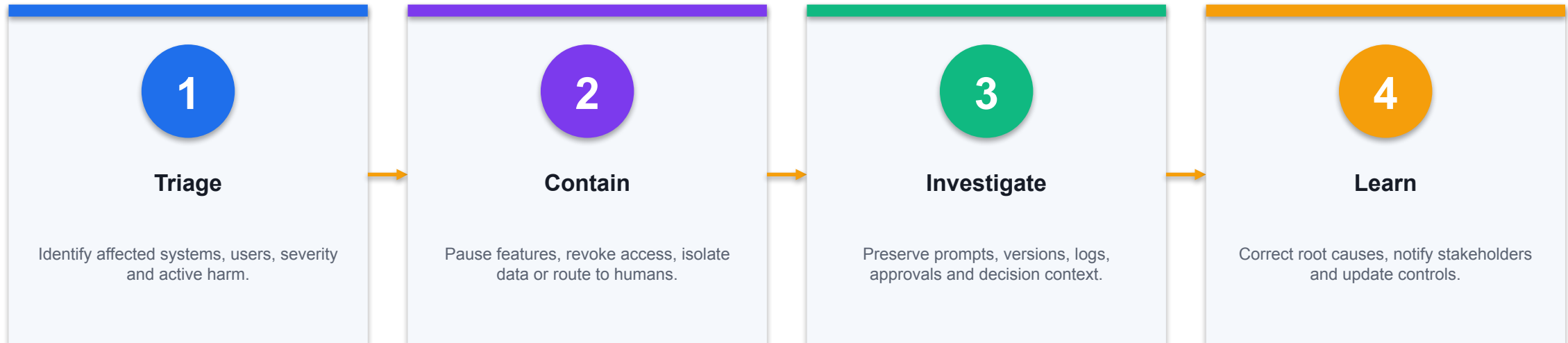
Real-world benefit, harm, complaints, appeals and override rates.

Controls

Guardrail coverage, reviewer quality, alert latency and remediation time.

A technically mature dashboard can still under-measure human outcomes and reviewer effectiveness.

AI incident response



AI incidents demand both technical response and a human-impact response.

Change and retirement

1 Triggers

New model, data, prompt, vendor, use, market or legal requirement.

MODULE 4

2 Revalidation

Re-run tests proportional to the change and its risk.

MODULE 4

3 Rollback

Keep a tested route to the last approved safe version.

MODULE 4

4 Retirement

Archive evidence, delete data appropriately and inform dependent users.

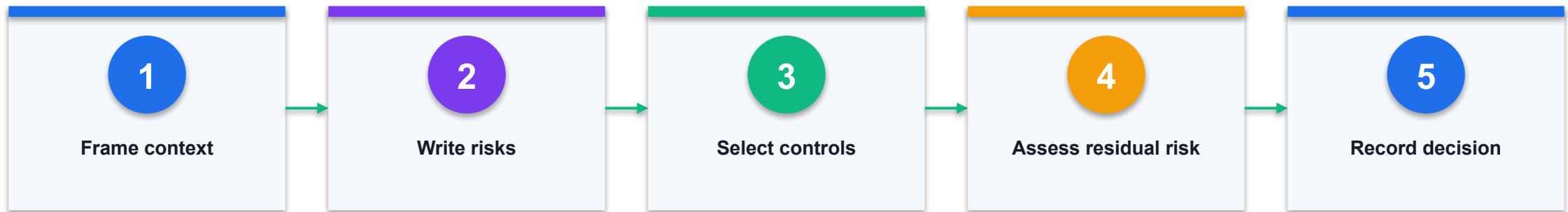
MODULE 4

THE POINT

A material change creates a new risk decision; version numbers alone do not preserve assurance.

Complete an AI Impact Assessment and Risk Register

Apply the governance lifecycle to assess impact, select controls and make a recorded go/no-go decision (A1, A3).



YOU'LL PRODUCE AI impact assessment, risk register and stage-gate decision

Tools: Prompt PDF, impact template, risk register CSV · Detailed procedure: Learner Guide, Activity 4

Verify · Complete an AI Impact Assessment and Risk Register

PASS WHEN

The assessment connects context, harms, controls, evidence, residual risk and an accountable decision with conditions.



Alternatives considered



Six specific risk statements



Controls cover four modes



Residual risk is rescored



Decision and review date recorded

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

COURSE PAUSE

00

End of Day 1

Review and reflect

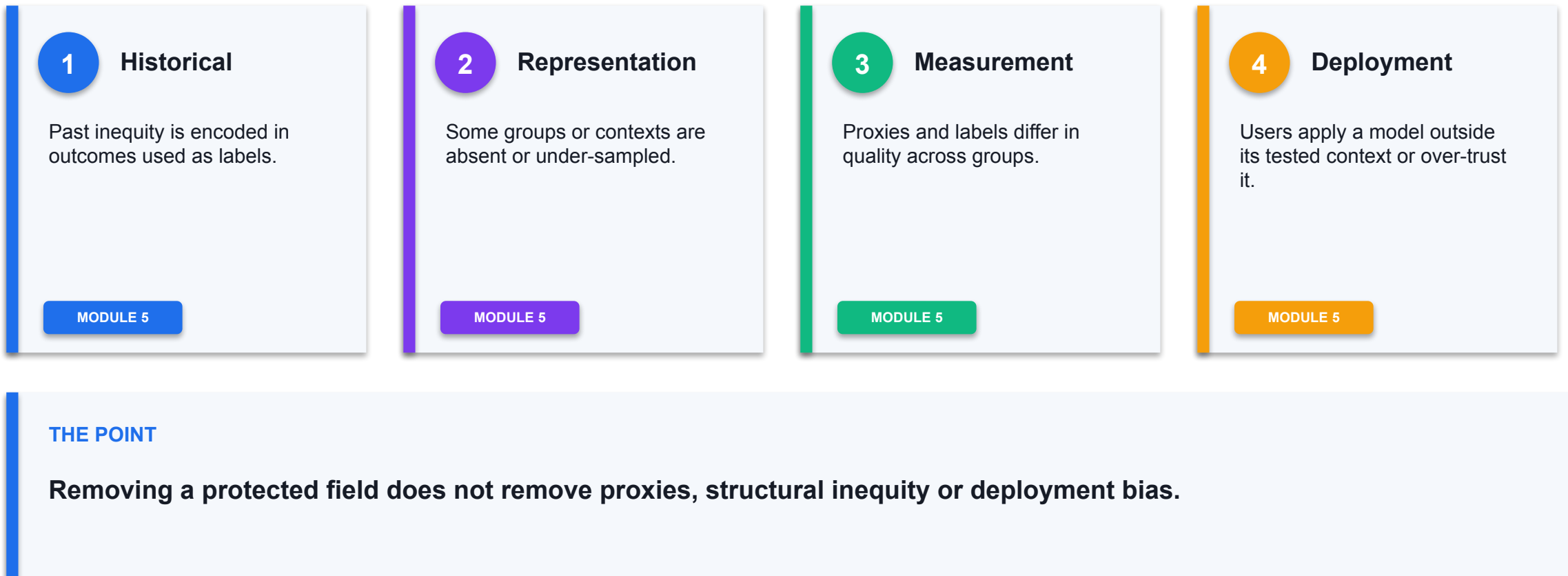
MODULE 5 · K3 · K4 · A1

05

Fairness and Explainability

Measure disparities and explain decisions without false certainty

Bias enters at every stage



Read a confusion matrix

1 True positive

A positive case is correctly identified.

MODULE 5

2 False positive

A negative case is incorrectly flagged positive.

MODULE 5

3 True negative

A negative case is correctly identified.

MODULE 5

4 False negative

A positive case is missed—often the costliest error.

MODULE 5

THE POINT

Fairness analysis begins by asking which error creates harm for whom.

Fairness metrics

1 Demographic parity

Positive decision rates are comparable across groups.

MODULE 5

2 Equal opportunity

True-positive rates are comparable across groups.

MODULE 5

3 Equalised odds

Both true-positive and false-positive rates are comparable.

MODULE 5

4 Calibration

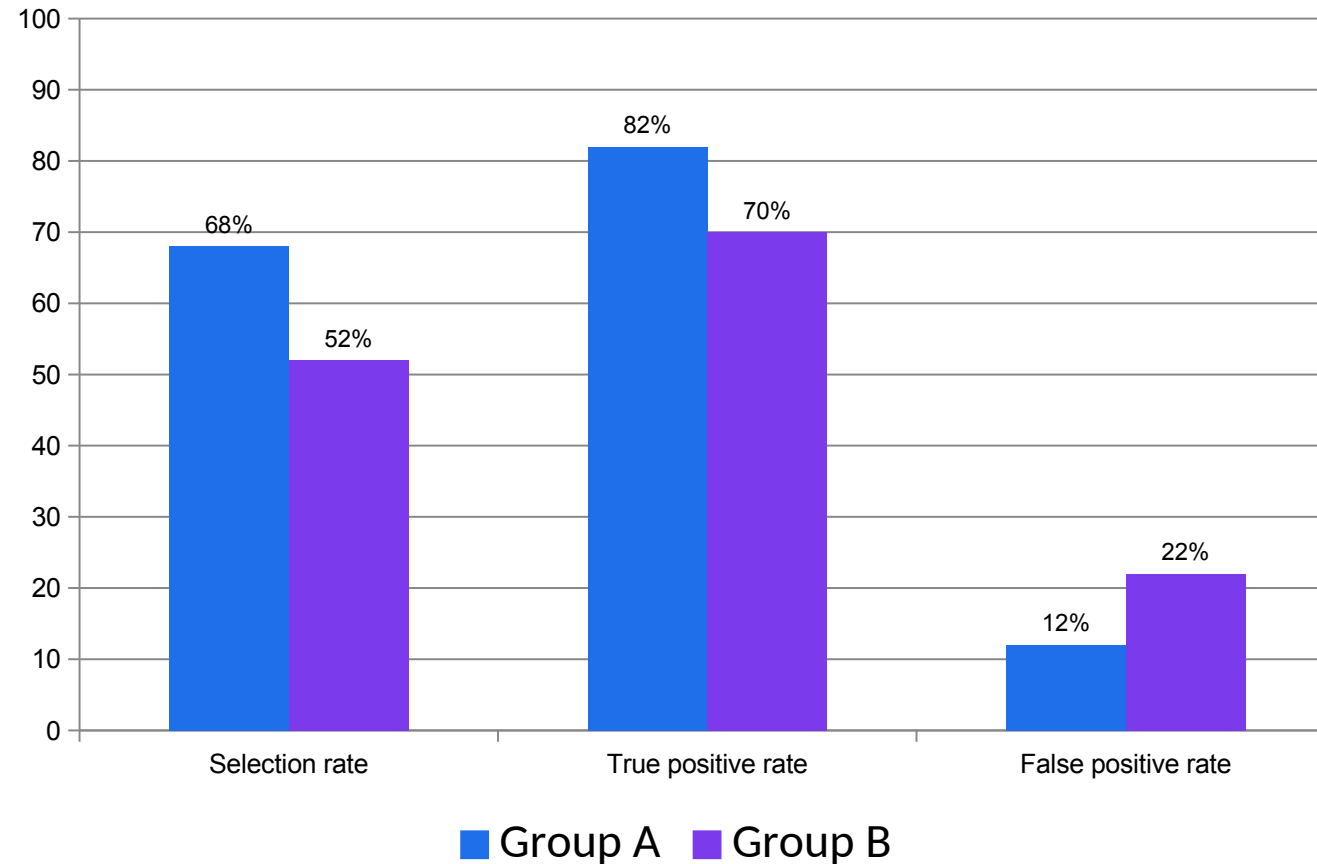
A given score has comparable meaning across groups.

MODULE 5

THE POINT

Metrics can conflict; select and justify the one aligned to the use case and harm model.

Fairness dashboard



Selection rate

Group A 68% versus Group B 52% indicates outcome disparity.

True-positive rate

Group A 82% versus Group B 70% indicates missed opportunity.

False-positive rate

Group A 12% versus Group B 22% indicates excess burden.

Response

Investigate data, thresholds, features and process—not just the metric.

Group B has a lower selection and true-positive rate, but a higher false-positive rate.

White-box and black-box explanations

1 White-box

Rules, linear models and small trees can be interpreted directly.

MODULE 5

2 Black-box

Complex ensembles and deep models usually need post-hoc methods.

MODULE 5

3 Global

Describe overall behaviour, influential features and population patterns.

MODULE 5

4 Local

Explain why one prediction changed for one instance.

MODULE 5

THE POINT

Model performance and interpretability are not a binary choice; choose evidence proportional to risk.

LIME, SHAP and PDP

1 LIME

Fits a simple local surrogate around one prediction.

MODULE 5

2 SHAP

Allocates contribution using Shapley-value reasoning.

MODULE 5

3 PDP

Shows the average marginal relationship between a feature and prediction.

MODULE 5

4 Caution

Correlation, unstable explanations and feature dependence can mislead.

MODULE 5

THE POINT

Explainability methods reveal model behaviour; they do not prove causality or fairness.

Counterfactual explanations

1 Question

What minimal change would alter this outcome?

MODULE 5

2 Use

Give an affected person a concrete, actionable reason.

MODULE 5

3 Constraint

Changes must be feasible, lawful and not shift burden unfairly.

MODULE 5

4 Risk

A technically valid counterfactual may recommend an impossible life change.

MODULE 5

THE POINT

A useful explanation respects human agency and the real-world ability to act.

Documentation for explainability

1 Model card

Intended use, performance, limits, groups and evaluation results.

MODULE 5

2 Data sheet

Provenance, consent, composition, processing and known gaps.

MODULE 5

3 System card

End-to-end behaviour, safeguards, tests and deployment context.

MODULE 5

4 Decision notice

Plain-language reason, human contact and contest route.

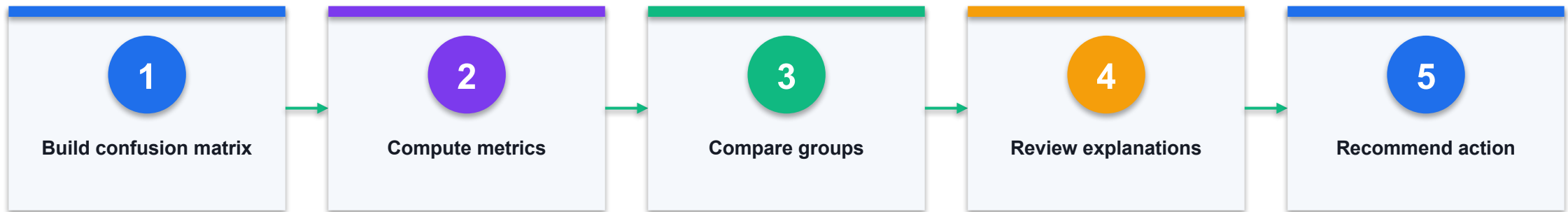
MODULE 5

THE POINT

Match documentation to audience while keeping every claim traceable to evidence.

Audit Fairness and Explainability

Compute group-level fairness indicators, interpret explanations and recommend proportionate mitigations (A1, A2).



YOU'LL PRODUCE Fairness scorecard and explanation review

Tools: Spreadsheet, fairness-data.csv, explanation samples · Detailed procedure: Learner Guide, Activity 5

Verify · Audit Fairness and Explainability

PASS WHEN

The scorecard shows correct metrics by group, interprets harm cautiously and connects mitigation to an owner and monitoring rule.



Confusion matrices
reconcile



At least four metrics per
group



Metric choice is justified



No causal claim from
correlation



Explanation supports
contestability

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

MODULE 6 · K4 · A2

06

Data Governance and Privacy

Make provenance, quality, access and retention governable

Data governance for AI

1 Ownership

Data owner approves purpose, access, quality thresholds and retention.

MODULE 6

2 Provenance

Record source, collection method, licence, consent and transformations.

MODULE 6

3 Quality

Measure completeness, accuracy, timeliness, consistency and representativeness.

MODULE 6

4 Control

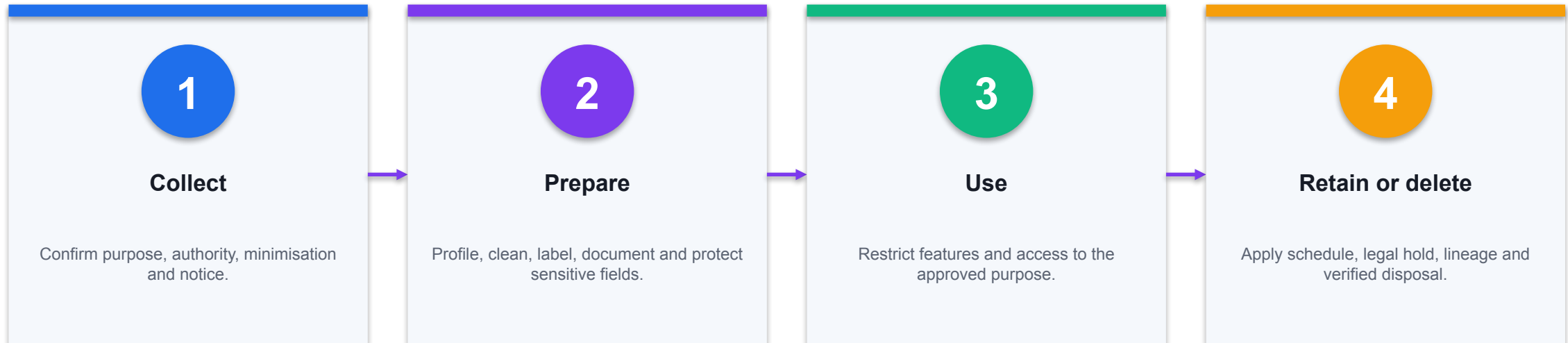
Enforce classification, access, sharing, retention and deletion.

MODULE 6

THE POINT

A model is only as governable as the evidence chain behind its data.

AI data lifecycle



Each hand-off needs a recorded owner, allowed purpose and control state.

Data lineage

1 Source

Where did the data originate and under what authority?

MODULE 6

2 Transform

Which joins, filters, labels and feature engineering changed it?

MODULE 6

3 Model

Which dataset version trained or evaluated which model version?

MODULE 6

4 Outcome

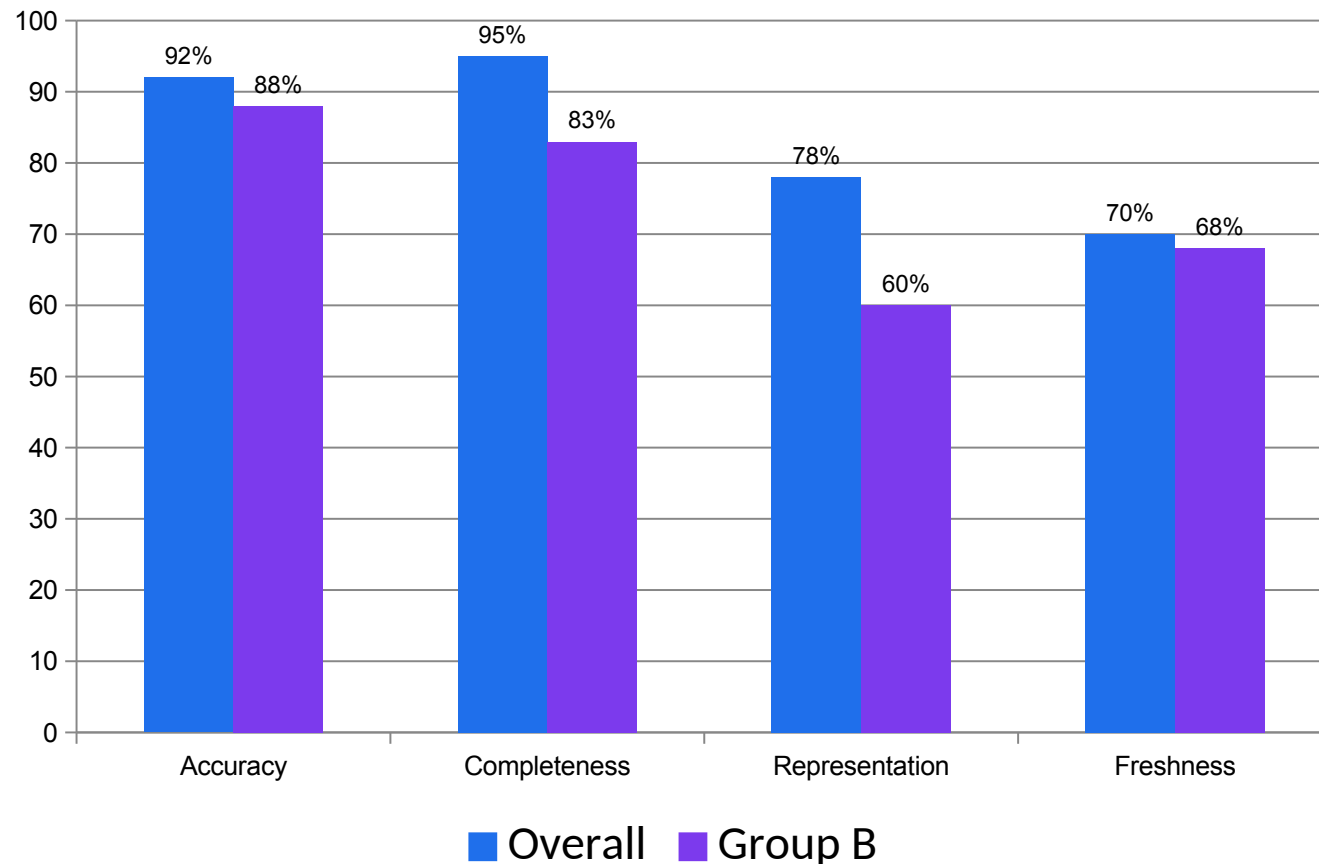
Which data and model version influenced a specific decision?

MODULE 6

THE POINT

Lineage turns an unrepeatable result into an auditable decision chain.

Data quality is contextual



Accuracy

Values reflect the real-world attribute closely enough for purpose.

Completeness

Missingness is measured overall and by relevant group.

Representativeness

Training data covers the population and conditions of use.

Freshness

Update frequency matches how quickly the phenomenon changes.

Aggregate quality can hide a material representation gap for a subgroup.

Privacy threat patterns

1 Linkage

Quasi-identifiers reconnect de-identified records to individuals.

MODULE 6

2 Membership inference

An attacker infers whether a person was in training data.

MODULE 6

3 Model inversion

Outputs reveal sensitive attributes or representative records.

MODULE 6

4 Prompt leakage

Users expose personal or confidential data to a generative system.

MODULE 6

THE POINT

De-identification reduces risk; it does not automatically make data anonymous or unrestricted.

Privacy-enhancing techniques

1 K-anonymity

Generalise quasi-identifiers so each record resembles at least $k-1$ others.

MODULE 6

2 Differential privacy

Bound an individual's influence using calibrated randomness.

MODULE 6

3 Federated learning

Train across locations without centralising raw data.

MODULE 6

4 Synthetic data

Generate representative records while testing disclosure and bias risk.

MODULE 6

THE POINT

Choose a technique by threat model; every privacy method introduces utility and implementation trade-offs.

Access, retention and deletion

1 Least privilege

Grant only the data and action needed for the role.

MODULE 6

2 Separation

Keep production data away from experimentation and public tools.

MODULE 6

3 Retention

Set period, trigger, owner and approved exceptions.

MODULE 6

4 Deletion

Remove copies, embeddings, caches and backups where required.

MODULE 6

THE POINT

A deleted source file may survive in a vector store, fine-tune, cache or derived feature table.

Data governance evidence

1 Data inventory

Dataset, owner, purpose, sensitivity, location and retention.

MODULE 6

2 Data sheet

Composition, collection, preprocessing, limitations and licences.

MODULE 6

3 Quality report

Thresholds, exceptions and group-level results.

MODULE 6

4 Access log

Who accessed which data, why, when and what changed.

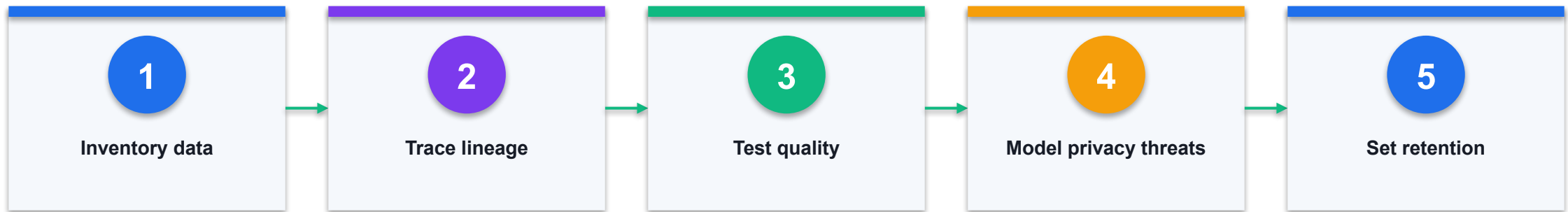
MODULE 6

THE POINT

Evidence should allow an independent reviewer to reconstruct the data decision.

Design Data Governance and Privacy Controls

Review AI data lineage, quality, privacy threats, access and retention controls (A2).



YOU'LL PRODUCE Data sheet, lineage map and privacy control plan

Tools: Prompt PDF, data inventory, lineage CSV, DPIA checklist · Detailed procedure: Learner Guide, Activity 6

Verify · Design Data Governance and Privacy Controls

PASS WHEN

The data plan can reconstruct provenance and sets measurable quality, access, privacy, retention and deletion requirements.



Purpose and authority
recorded



Lineage reaches model
decision



Quality measured by group



PET maps to threat



Derived copies included in
deletion

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

COURSE PAUSE

00

Lunch Break

45 minutes

MODULE 7 · K4 · A2

07

Generative AI Security and Copyright

Secure the data-model-application-human stack

A Layered Generative AI Security Model



Data

Provenance, access, poisoning and privacy.

Model

Provider, weights, prompts and evaluation.

Application

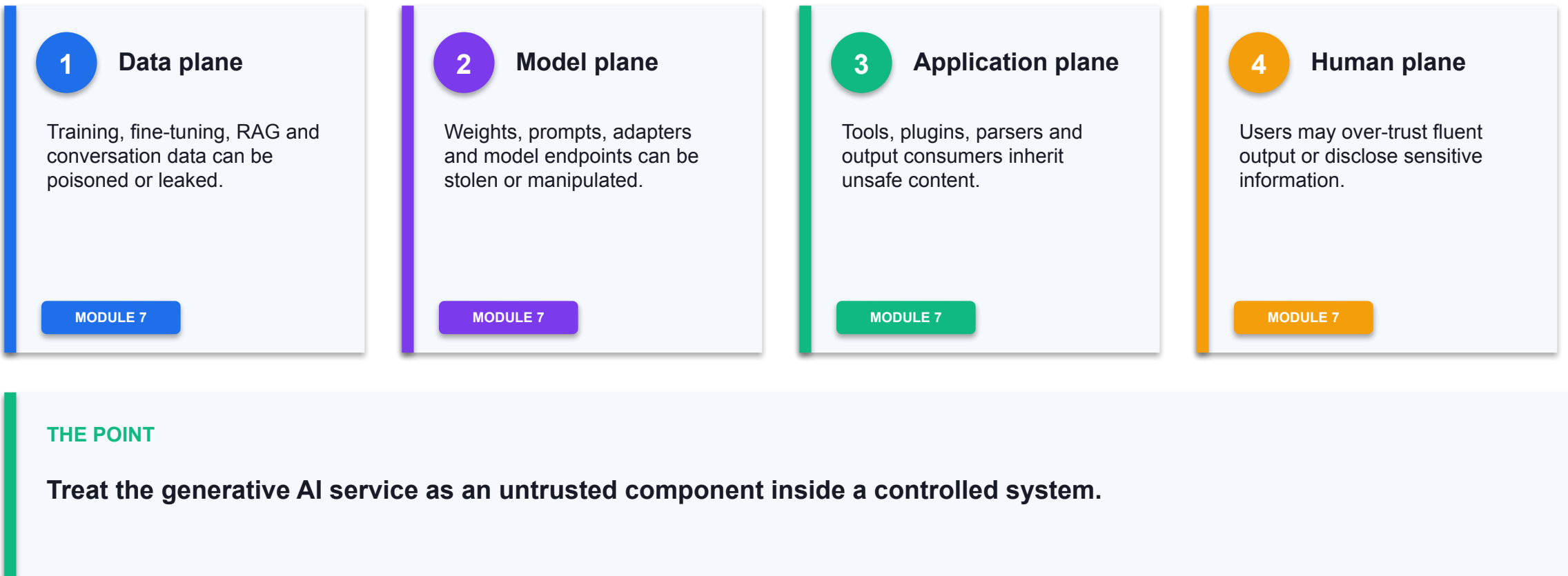
Retrieval, tools and output handling.

Human

Oversight, use policy and incident response.

Assume every layer can fail; design overlapping controls.

Generative AI expands the attack surface



OWASP LLM Top 10: input and data

1 Prompt injection

Direct or indirect instructions override intended behaviour.

MODULE 7

2 Sensitive disclosure

The system reveals private, confidential or regulated information.

MODULE 7

3 Supply chain

Models, datasets, packages and providers introduce hidden risk.

MODULE 7

4 Data/model poisoning

Manipulated data or artefacts corrupt behaviour or create backdoors.

MODULE 7

THE POINT

Input filtering alone cannot solve prompt injection; constrain privileges, data and downstream actions.

OWASP LLM Top 10: output and agency

1

Improper output handling

Unsafe model output reaches code, queries, browsers or users.

MODULE 7

2

Excessive agency

The model can perform actions beyond minimum necessary privileges.

MODULE 7

3

System prompt leakage

Hidden instructions are exposed and mistaken for a security boundary.

MODULE 7

4

Unbounded consumption

Inputs or loops exhaust cost, compute, time or service capacity.

MODULE 7

THE POINT

Validate model output at the trust boundary and require approval for high-impact actions.

RAG and vector-store risks

1 Poisoned documents

Malicious content steers retrieval or injects hidden instructions.

MODULE 7

2 Access bleed

Chunks from one user, tenant or classification reach another.

MODULE 7

3 Embedding leakage

Vector representations can expose sensitive relationships.

MODULE 7

4 Stale knowledge

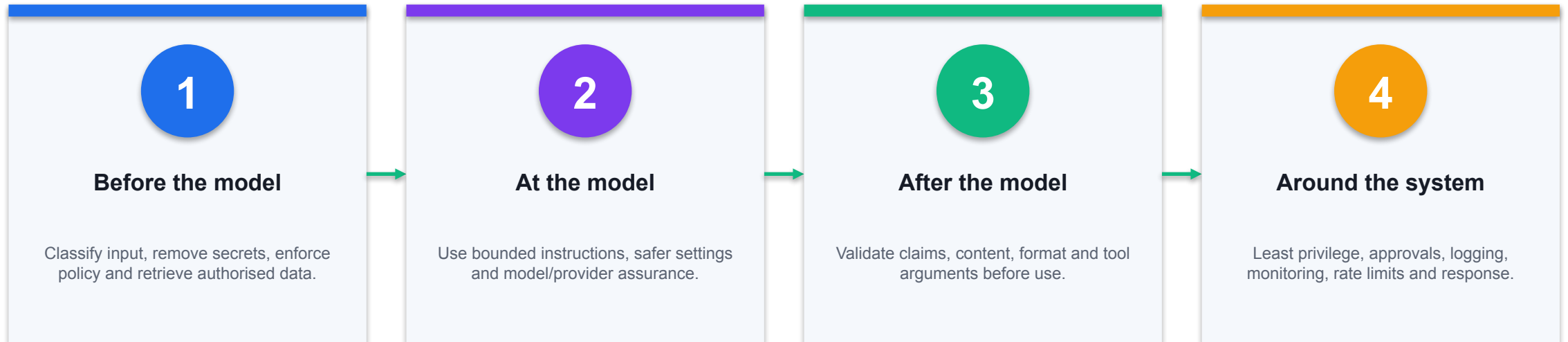
Outdated or revoked documents remain retrievable.

MODULE 7

THE POINT

Carry identity, permissions, provenance and expiry from source document to every chunk and retrieval.

Layered generative AI controls



Defence in depth assumes every individual guardrail will eventually fail.

Hallucination and misinformation

1 Cause

The model generates probable language, not verified truth.

MODULE 7

2 Exposure

Open-ended questions and missing evidence increase uncertainty.

MODULE 7

3 Control

Ground responses, cite sources, verify claims and calibrate refusal.

MODULE 7

4 Oversight

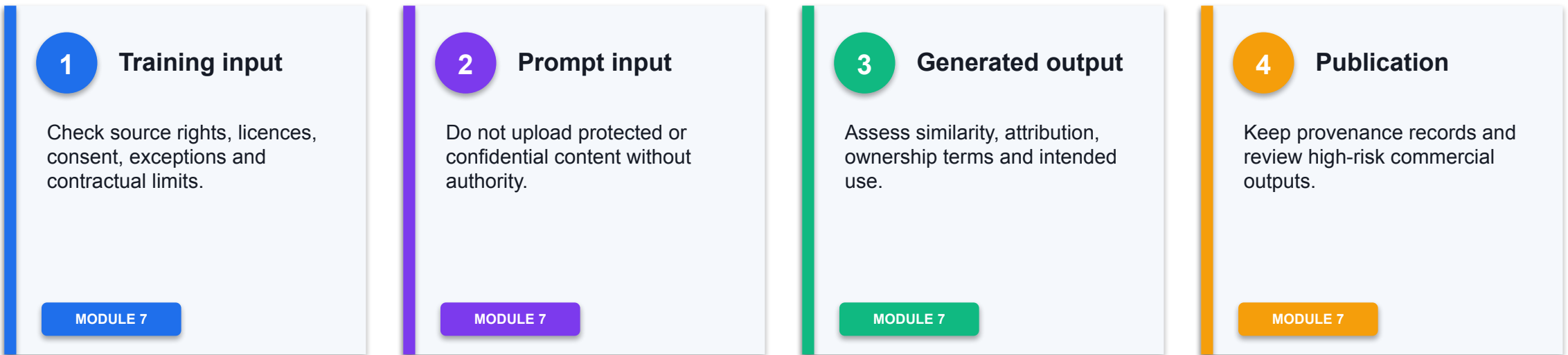
Route consequential content to qualified human review.

MODULE 7

THE POINT

Fluency is not evidence; require provenance for claims that drive decisions.

Copyright across the GenAI lifecycle



THE POINT

Copyright risk depends on jurisdiction and facts; governance should preserve provenance and enable review.

Security architecture evidence

1 Threat model

Assets, actors, trust boundaries, abuse cases and mitigations.

MODULE 7

2 Evaluation

Prompt-attack, leakage, output, access and cost tests.

MODULE 7

3 Telemetry

Prompt category, retrieval source, tool call, approval and outcome logs.

MODULE 7

4 Runbook

Containment, key rotation, corpus purge, rollback and notification.

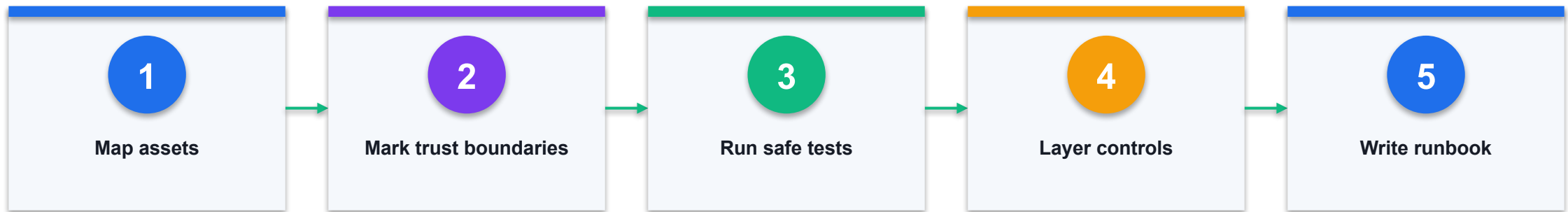
MODULE 7

THE POINT

A secure design can show which layer stops each threat and what evidence proves it.

Threat-Model a Generative AI Assistant

Apply layered generative AI security, privacy and copyright controls (A2).



YOU'LL PRODUCE Threat model, adversarial prompt results and control architecture

Tools: Prompt PDF, architecture.json, attack prompts, control checklist · Detailed procedure: Learner Guide, Activity 7

Verify · Threat-Model a Generative AI Assistant

PASS WHEN

The threat model covers data, model, application and human layers; every high-risk abuse case has multiple controls and evidence.



Trust boundaries drawn



OWASP risks mapped



No real secrets used



Tool permissions are least
privilege



Copyright and incident
controls included

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

COURSE PAUSE

00

Tea Break

10 minutes

MODULE 8 · A2 · A3

08

Human Oversight and Accountability

Keep meaningful human control where consequences matter

Meaningful Human Oversight



Understand

Reviewer sees evidence and uncertainty.

Challenge

Reviewer can disagree without friction.

Intervene

Reviewer can pause, override or escalate.

Learn

Overrides and outcomes improve the system.

Human presence matters only when it changes decisions and outcomes.

Human involvement models

1 Human-in-the-loop

A person approves each consequential recommendation before action.

MODULE 8

2 Human-on-the-loop

AI acts within bounds while a person supervises and can intervene.

MODULE 8

3 Human-out-of-the-loop

AI acts without real-time review; reserve for low-risk, reversible tasks.

MODULE 8

4 Human-in-command

People set goals, limits, authority, escalation and shutdown conditions.

MODULE 8

THE POINT

Meaningful oversight depends on power, information, time and competence—not the presence of a checkbox.

Choose oversight by risk

1 Severity

Greater potential harm requires stronger review and authority.

MODULE 8

2 Reversibility

Irreversible outcomes require intervention before action.

MODULE 8

3 Uncertainty

Novel contexts and weak confidence increase escalation.

MODULE 8

4 Scale

High-volume decisions need sampling, monitoring and systemic safeguards.

MODULE 8

THE POINT

Automate low-risk routine decisions; preserve human approval for high-impact exceptions.

Design a meaningful review

1 Information

Show evidence, uncertainty, alternatives and policy—not only a score.

MODULE 8

2 Time

Give the reviewer enough time to understand and challenge the result.

MODULE 8

3 Authority

Allow override, pause, escalation and correction without penalty.

MODULE 8

4 Competence

Train reviewers and monitor the quality of their decisions.

MODULE 8

THE POINT

A reviewer who cannot understand or change the decision is not an effective control.

Automation bias and rubber-stamping

1 Anchoring

The AI recommendation becomes the default starting point.

MODULE 8

2 Complacency

Repeated correct outputs reduce vigilance over time.

MODULE 8

3 Asymmetry

Overriding the model takes more effort or creates perceived blame.

MODULE 8

4 Countermeasure

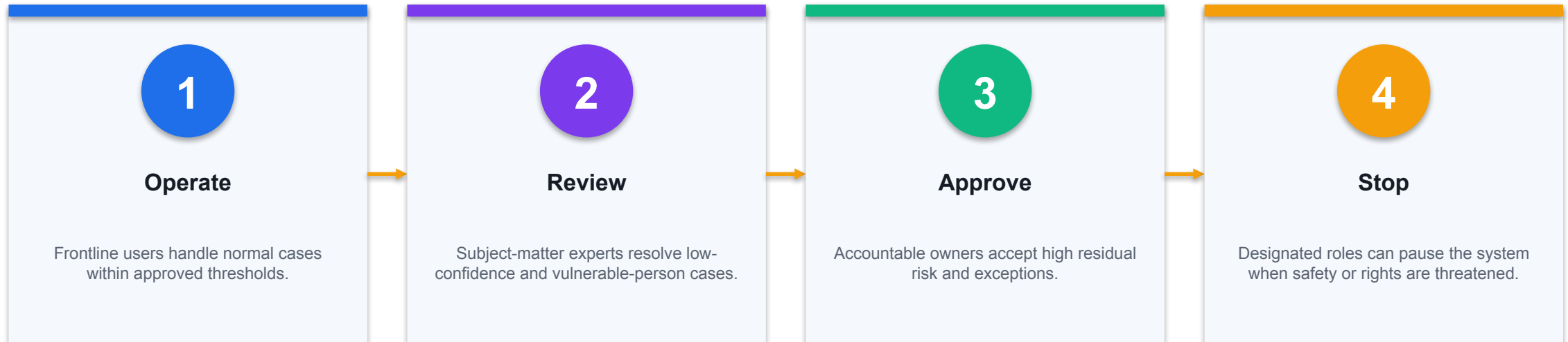
Hide score initially, require reasons, sample reviews and measure overrides.

MODULE 8

THE POINT

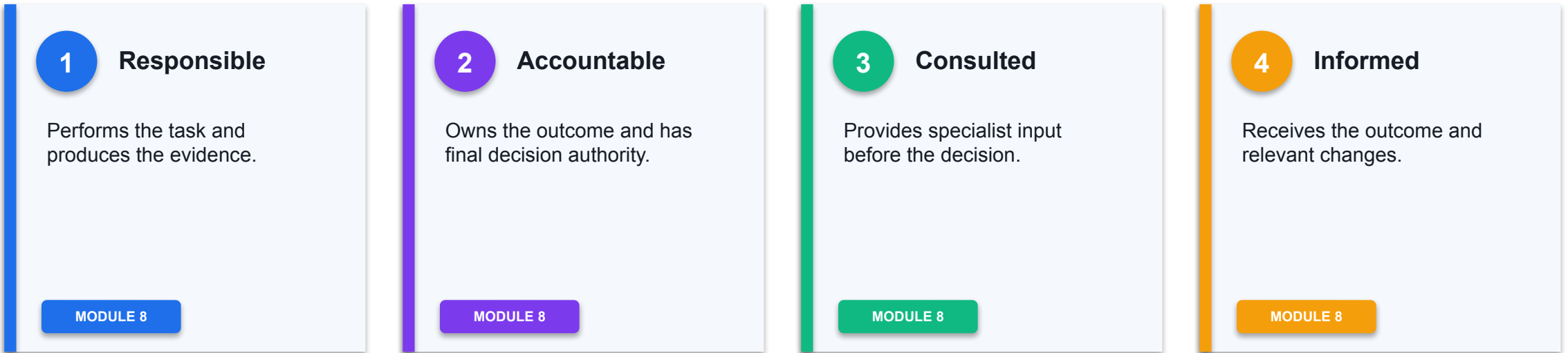
Human review can create false assurance unless the interface and incentives support genuine challenge.

Decision rights and escalation



Every boundary needs a trigger, response time, owner and fallback.

RACI for AI governance



THE POINT

There should be one accountable owner for each material risk decision.

Audit trail and contestability

1 Trace

Record input, model/data version, output, reviewer and final action.

MODULE 8

2 Explain

Provide a reason suited to the affected person's needs.

MODULE 8

3 Correct

Enable factual correction and rerun where appropriate.

MODULE 8

4 Appeal

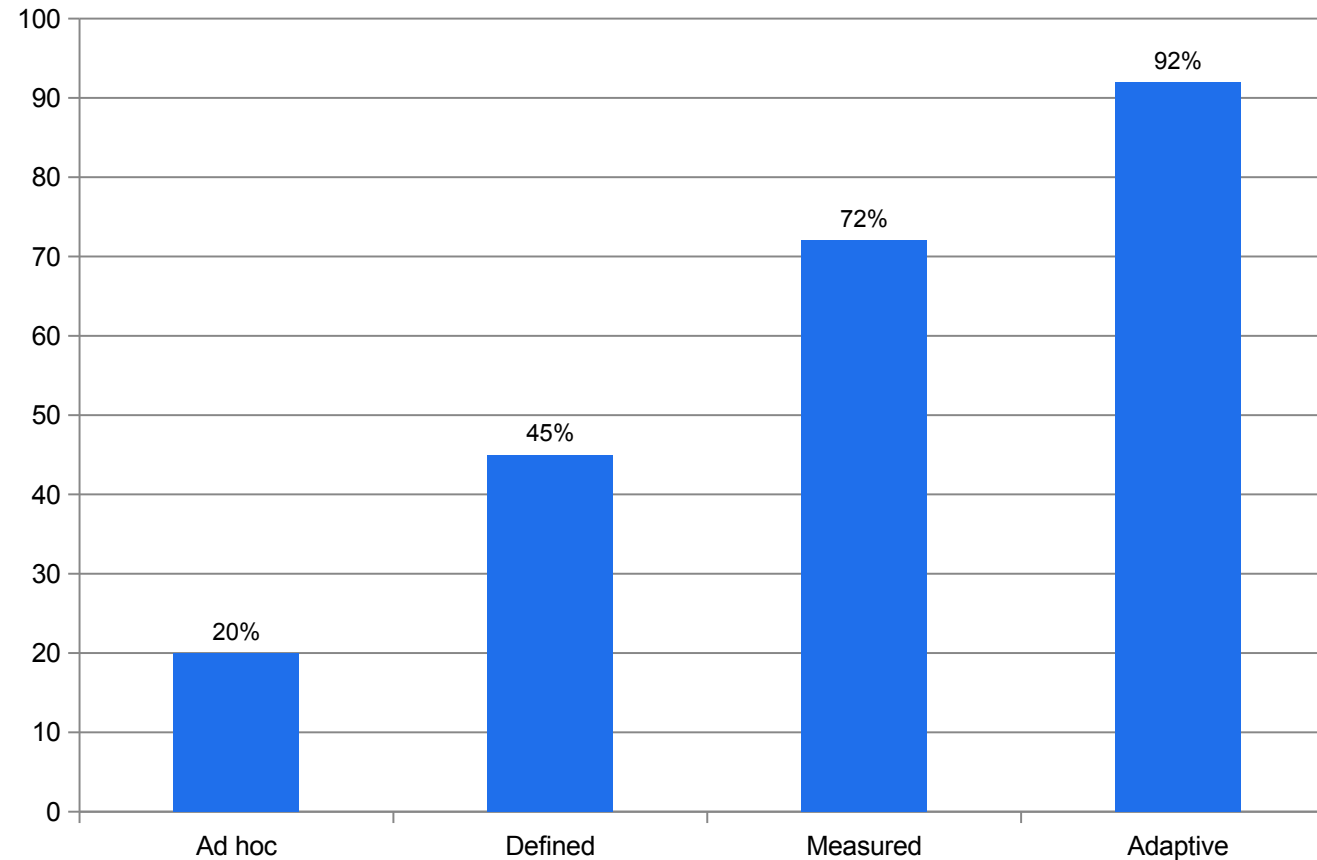
Offer timely independent review and communicate the outcome.

MODULE 8

THE POINT

Redress completes accountability by giving affected people a path from explanation to remedy.

Governance maturity roadmap



Ad hoc

Teams rely on individual judgement and scattered documents.

Defined

Inventory, policy, roles, risk tiers and minimum controls exist.

Measured

Control effectiveness, outcomes, incidents and fairness are monitored.

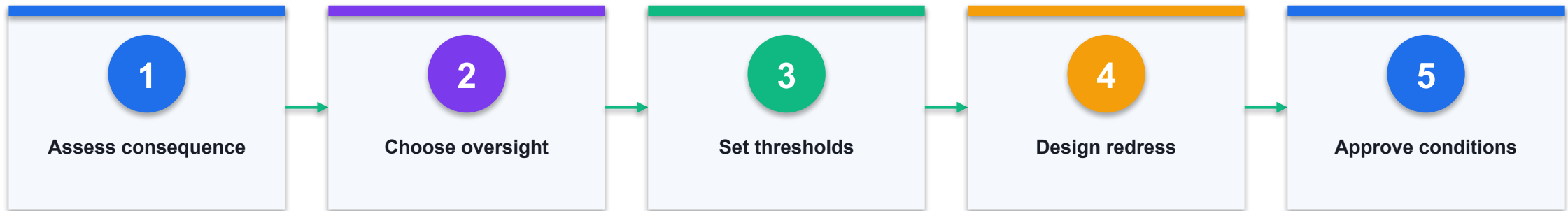
Adaptive

Evidence drives continuous improvement, challenge and transparent reporting.

The maturity gain comes from evidence, learning and outcomes—not from producing more policy documents.

Design Human Oversight and a Governance Decision

Select meaningful human involvement, escalation, monitoring and redress for an AI deployment (A2, A3).



YOU'LL PRODUCE Oversight matrix and governance board decision pack

Tools: Prompt PDF, decision scenarios, oversight matrix, board template · Detailed procedure: Learner Guide, Activity 8

Verify · Design Human Oversight and a Governance Decision

PASS WHEN

The oversight design is risk-proportionate, gives reviewers real authority, addresses automation bias and includes contestability and stop conditions.



Oversight choice is justified



Reviewer has time and authority



Escalation thresholds are measurable



Appeal and correction exist



Board decision records residual risk

If evidence is incomplete, record the gap, owner and due date—do not mark the control effective.

WRAP-UP

09

From Principles to Practice

Evidence · accountability · continuous improvement

What You Can Now Do

1 Identify

Recognise principles, stakeholders, harms and governance processes.

COURSE SUMMAR

2 Assess

Tier use cases and complete an AI impact and risk assessment.

COURSE SUMMAR

3 Control

Design fairness, data, security and oversight controls.

COURSE SUMMAR

4 Decide

Present evidence and make a recorded, accountable deployment decision.

COURSE SUMMAR

THE POINT

Responsible AI is a repeatable organisational capability, not a one-off compliance exercise.

Reference Frameworks

1 Singapore

Model AI Governance Framework · AI Verify · GenAI governance.

REFERENCES

2 NIST

AI RMF 1.0 · Generative AI Profile AI 600-1.

REFERENCES

3 ISO

ISO/IEC 42001 AI management system.

REFERENCES

4 OWASP

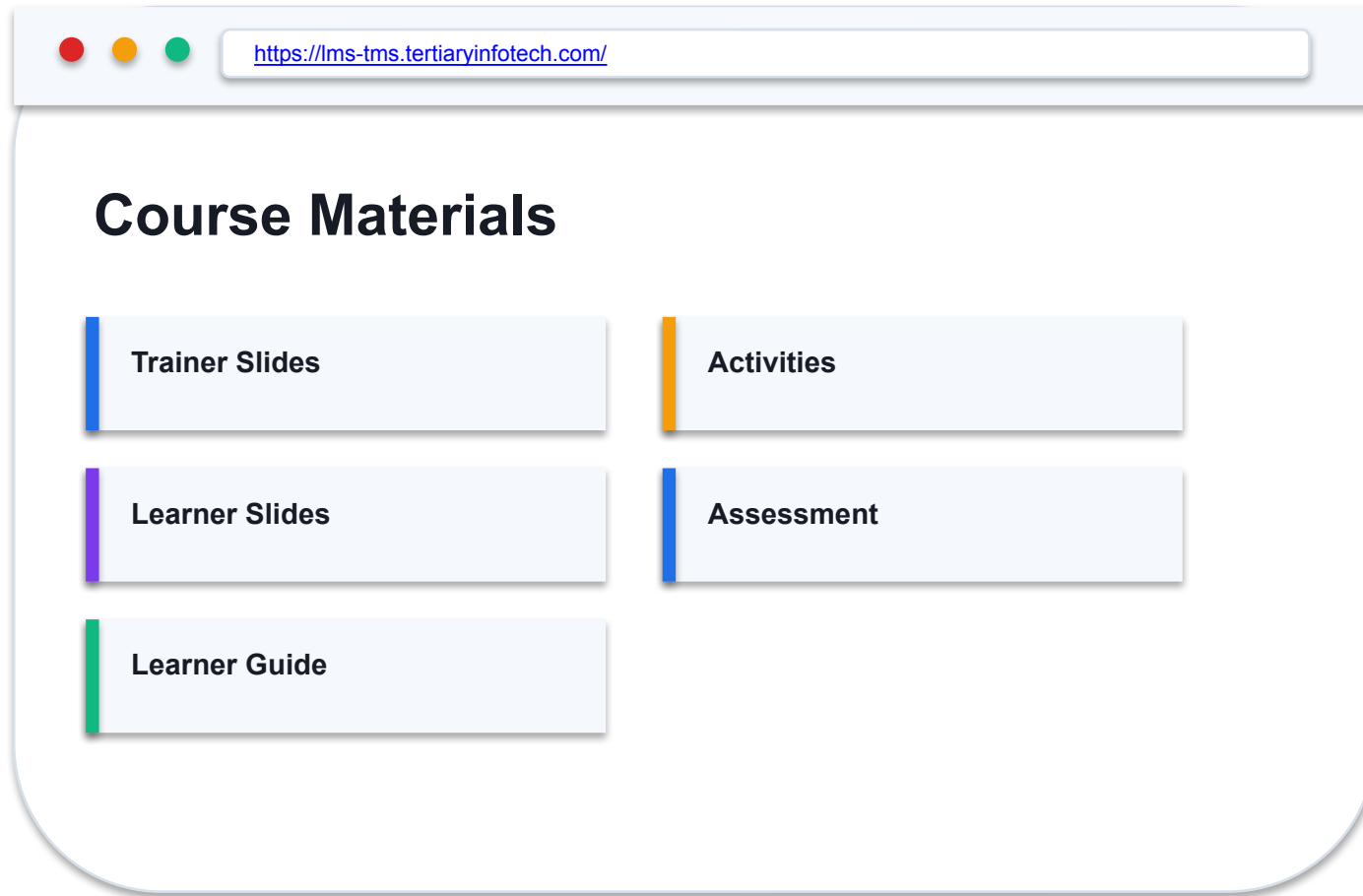
Top 10 for LLM Applications 2025.

REFERENCES

THE POINT

Use primary framework sources and document which version informed each decision.

Courseware and Assessment Hub



- 1 Approved courseware
- 2 Assessment attendance
- 3 Candidate question papers
- 4 Secure submission area

Assessment Reminder

1 WA

4 short-answer questions · K1–K4 · 1 hour.

ASSESSMENT

2 Case Study

3 scenario questions · A1–A3 · 1 hour.

ASSESSMENT

3 Submit

Upload completed question papers to the LMS.

ASSESSMENT

4 Protect

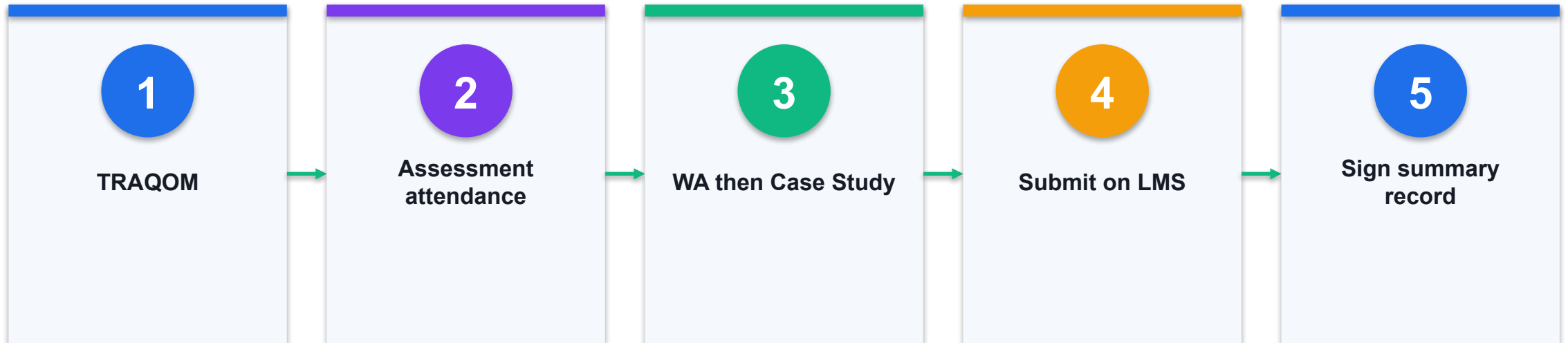
Do not photograph or distribute assessment content.

ASSESSMENT

THE POINT

Answer keys remain trainer-only and are never attached to learner-facing systems.

Assessment Flow



Complete the steps in sequence and confirm each submission.

Digital Attendance (Mandatory)

1 AM attendance

Scan the SSG digital-attendance QR code when the trainer displays it.

TRAQOM

2 PM attendance

Repeat after lunch; confirm the submission screen before closing.

TRAQOM

3 Assessment attendance

Complete the separate assessment attendance at the scheduled time.

TRAQOM

4 Funding requirement

Maintain at least 75% attendance and complete all prescribed assessments.

TRAQOM

THE POINT

TRAQOM attendance is a formal funding record—check each submission before continuing.

COURSE COMPLETE

Thank You

Build AI systems people can understand, challenge and trust.

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